Subsidizing Crises: Evidence from Norway's Electricity Market*

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October 14, 2026

Abstract

We examine consumer responses to the 2022 European Energy Crisis, using Norway's zonal electricity market and the introduction of a subsidy policy as a natural experiment. Because the policy subsidized a fixed percentage of electricity costs strictly above a pre-established threshold, it created a kink in the consumer's marginal price schedule. Exploiting this change in slope and using high-frequency data, we estimate local demand responsiveness at the point where the subsidy becomes binding. We find price elasticities between -1.1 and -0.5, which are substantially larger in absolute value than the historical demand elasticities for Norway. Employing administrative consumption records from over 1.5 million households, we show that zones more interconnected with other countries experienced a sustained reduction in household electricity consumption relative to both less interconnected zones and to the pre-crisis and pre-subsidy period.

JEL codes: L51, Q41, Q48, H31

Keywords: consumer behavior, electricity markets, household electricity consumption, subsidies, government support policies.

*We thank Gordon Leslie from Monash University, Lassi Ahlvik, Olvar Bergland and Knut Einar Rosendahl from Norwegian University of Life Sciences (NMBU), seminar participants at the Friedrich-Alexander-Universität, University of Umeå, the Finance and business analytics group at NMBU, two anonymous referees, and the Editor for their helpful comments.

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1 Introduction

The European Energy Crisis began with a tightening of energy markets toward the end of 2021 and intensified substantially during 2022, with elevated market tensions persisting into 2023 before easing around the middle of that same year. In particular, the crisis caused an abnormal surge in wholesale electricity prices in almost all European markets.[1] Two main factors that contributed to the spike in prices were the reduced gas flows to Europe from Gazprom and that France became a net importer of electricity, drawing power from neighbors and adding massive demand to an already tight market.[2,3] Russia's full-scale invasion of Ukraine in February 2022 further exacerbated these developments, particularly through its effects on European gas supplies and energy prices.

In this paper, we study a subsidy policy implemented in December 2021 to help consumers with the rising costs of electricity in Norway. The external supply shock from the crisis was transmitted through the electricity network of several countries. In this context, we address two main questions. First, how responsive are consumers to electricity prices that substantially exceed historical averages? Second, what effect do net price increases have on electricity consumption? We shed light on these questions by estimating the price elasticity of electricity, exploiting both a change of slope in the subsidy's design and spatial variation in the interconnectivity of Norway's electricity pricing zones.

Our contribution consists of analyzing a distinct policy mechanism designed to mitigate the crisis' impact. The Norwegian subsidy program has no direct equivalent in other European markets. Unlike interventions that capped wholesale prices or targeted the generation sector, the Norwegian model directly subsidized retail prices for consumers while partially preserving marginal price signals. This provides a novel setting for evaluating the trade-off between shielding households from price shocks and maintaining incentives for conservation.

We use data on wholesale electricity prices and weather conditions for Norway's five bidding zones. Consumers in Norway are exposed to electricity prices that reflect real-time wholesale prices, thus we concentrate on the overall effect of those shocks on households consumption of electricity. We complement those data with administrative consumption

[1] See ECB (2022); Euronews (2025).

[2] Gazprom fulfilled long-term contracts but stopped selling spot gas (top-up supplies), likely to apply political pressure regarding the Nord Stream 2 pipeline. For a study on these supply cuts see Di Bella et al. (2024).

[3] France became a net importer of electricity for the first time in 30 years. See www.banque-france.fr.

records from over 1.5 million households in Norway and from January 2021 through December 2023. This dataset also contains household income, the surface of the dwelling, the number of rooms, and the building's year of construction.

The subsidy was implemented in the form of a one-sided subsidy-for-differences on consumption to compensate for the high-price levels. In other words, if the electricity spot price reached a level above a pre-established threshold, then the government subsidized a fixed percentage of the portion of the price above the threshold. This created a control-treatment environment in which consumers can be in the treatment group when electricity prices are above the pre-established threshold (therefore, activating the subsidy) and consumers belong to the control group when the price is below that same threshold. We estimate this treatment effect using a kink regression design model (Card et al. (2015)). This method gives us estimates on the price elasticity of demand at the price threshold. At that point, we find that demand for electricity became less inelastic with the introduction of the subsidy.[4] Two hypotheses come to mind: consumers perhaps were unaware of the subsidy and they simply react to higher-than-average prices, or consumers are adapting to the new price regime because they know that it is unclear whether the subsidy will be permanent or not. Based on anecdotal evidence from consumers in Norway, we think the most likely explanation is due to the latter.

Before electricity prices started to climb, demand elasticities were between -0.5 and -0.1. Following the beginning of the crisis in the fall of 2021, average household electricity consumption decreased. This reduction continued after the subsidy was introduced in December 2021, despite the policy covering between 80 and 90% of the price above the threshold. However, even though there was a price subsidy in place, consumers were exposed to relatively high net prices, at which our elasticity estimates indicate greater demand responsiveness than at pre-crisis price levels. When calculating elasticities using traditional methods such as log-log regressions with instrumental variables and the net price as the independent variable, we find elasticities twice as large as models using raw prices, but they still suggest demand is highly inelastic because they ignore the change-in-slope of the net price. That occurs because a traditional instrumental variable regression does not take into account the actual structure

[4]We also estimated price elasticities using a simple instrumental variable log-log regression. We find that demand is highly inelastic in some areas of the country before the policy was put in place. This is consistent with recent findings by Hofmann and Lindberg (2019) and is due to the high degree of electrification in Norway and thus the lack of energy substitutes for basic needs such as heating and common transportation needs such as electric vehicles. See Reiss and White (2005) for a general treatment of the estimation of elasticities in electricity markets.

of the subsidy. The use of the regression kink design explicitly takes this into account. Using this method, we find elasticities around the threshold of between -1.1 and -0.5, depending on the specific subsidy phase. At the same time, after-subsidy prices increased on average by 37% relative to the pre-subsidy period in the zones with high exposure to the supply shock, which together with the elasticity values obtained, the predicted consumption drop in electricity is closer in magnitude to the actual drop in demand than if we use historical demand elasticities for Norway.

We assess the effect of net price increases on electricity consumption through a dynamic differences-in-differences model to estimate an average treatment on the treated effect using the bidding zones that are the least connected to the rest of the European markets as a control group and the start of the subsidy policy as the event time. Norway offers a unique setup because its market is segmented by zones—zonal pricing—and their respective levels of interconnectivity to the rest of Europe largely differ, providing a control-treatment environment within the *same* market.[5] This addresses potential concerns regarding heterogeneous preferences. While this estimate does not capture a causal effect of the subsidy alone since electricity prices were on the rise independently of the government policy, it provides the average treatment effect on the treated of the net price and allows to quantify its magnitude and duration. We observe a sustained decrease in consumption in the eight months after the policy began relative to the control zones and to the pre-crisis and pre-subsidy period.

Leveraging the rich microdata on household characteristics, we estimate the heterogeneous effects of the net price changes using a differences-in-differences model with heterogeneity. The coefficient of interest identifies the differential treatment effect relative to a baseline category within the treated group. We find that treatment responses increase systematically with dwelling size and the number of rooms but are attenuated in newer buildings, a pattern consistent with differences in insulation quality and heating efficiency. Finally, while the policy affects electricity consumption for all households, the magnitude of this effect is significantly larger for higher-income households.

We conclude by discussing the consequences of the current subsidy policy relative to a fixed-price regime known as 'Norgespris', which was introduced in October 2025. We discuss how with the high price levels characterizing the early crisis period, a fixed-price policy would incur significantly higher total costs than the subsidy program.

[5] As explained later in the paper, the treated group has a total of 5.2 GW of interconnection capacity with the rest of Europe, whereas the control group has virtually zero.

Relations to the literature. Our contributions to the literature can be grouped into three main categories.

First, we contribute to the rapidly growing literature on household responses to the European Energy Crisis. Recent studies document significant reductions in energy consumption driven by price shocks and tariff changes (e.g., Ruhnau et al. (2023), Ajayi et al. (2024), Ahlvik et al. (2025), and Garnache et al. (2025)), alongside shifts in long-term cost expectations (Singhal et al. (2026)). While these studies report demand reductions ranging from 10 to 23%, Alpino et al. (2024) find highly inelastic responses among industrial consumers. Related research examines the crisis's broader market consequences. Brolinson et al. (2026) document its effects on housing prices and the relative valuation of energy-efficient homes, while Kröger et al. (2025) examine changes in long-term energy contracting and firms' hedging behavior. We extend this literature by estimating the local price elasticity of household electricity demand induced by a unique, large-scale subsidy program.

Second, our findings inform the broader literature on behavioral adaptations and policy design in energy markets. Since electricity pricing in our setting is based on hourly rates rather than increasing-block pricing, complications regarding marginal versus average prices (e.g., Ito (2014)) do not directly apply to our findings. Recent work emphasizes the role of non-price interventions, such as public appeals (Reiss and White (2008), Brewer and Crozier (2025)), financial rewards (Amberg et al. (2025)), and crisis-induced "mindset shifts" (Dertwinkel-Kalt et al. (2024)). Our results provide empirical support for these behavioral mechanisms: Norwegian households adapted to the high-price environment rather than reacting conventionally to the immediate price relief. Furthermore, compared to policies targeting the generation sector like the "Iberian solution" (Fabra and Montero (2023)), the Norwegian subsidy preserves a kinked marginal price signal, which we explicitly model using a regression kink design (Card et al. (2015)) to estimate the demand elasticity exactly at the policy threshold.

Finally, we contribute to the literature on the economics of electricity transmission. Over the past few decades, there has been a trend towards increasing interconnectivity between different electricity markets to fully exploit complementarities in generation portfolios. This has been particularly evident in Europe, although several other jurisdictions have witnessed similar changes (Gonzales et al. (2023), Hausman (2024)). However, this increased interconnectivity exposes broader market regions to the same supply and demand shocks, potentially

dampening the benefits of a diversified market reach.[6] Our consumption results provide additional evidence on the consequences of differing levels of network integration in regions subject to the same national policy framework.

2 Regulatory and Historical Background

Norway has historically had comparatively low electricity prices, especially compared to other European countries. This favorable situation is primarily attributable to the fact that electricity is sourced by approximately 90% hydro and the rest by wind and fossil-fueled power plants.[7] However, there are significant regional differences: electricity prices in Norway vary markedly between the north and the south, due to differences in supply, demand, and transmission capacity. To reflect these regional conditions, the electricity market is divided into five bidding zones (NO1–NO5), as shown in Figure 1. This zonal structure has been central to how recent price shocks were transmitted across the country.[8] During the time of our study, approximately 90% of households had contracts directly indexed to the day-ahead spot price, meaning that shocks in wholesale markets were almost immediately reflected in household bills.[9]

In October 2021, a sharp increase in wholesale electricity prices marked the beginning of an unprecedented price shock, which was further amplified in February–March 2022 following Russia's invasion of Ukraine and the resulting turbulence in European energy markets. The southern zones: NO1 (East), NO2 (South), and NO5 (West) were hit hardest. Their integration with continental markets through interconnectors to Denmark, Germany, the Netherlands, and the United Kingdom meant that domestic prices closely tracked the high levels abroad. At the same time, low water levels in southern hydropower reservoirs constrained supply, further fueling the surge.

By contrast, zones NO3 (Central) and NO4 (North) were largely shielded from the crisis. Their reservoirs were relatively well supplied, and crucially, these zones are not directly connected to continental Europe or the United Kingdom. In contrast, Southern Norway (NO1, NO2, and NO5) is tightly integrated with external electricity markets through a set of large subsea interconnectors, including Skagerrak 1–4 to Denmark (1,700 MW), NorNed

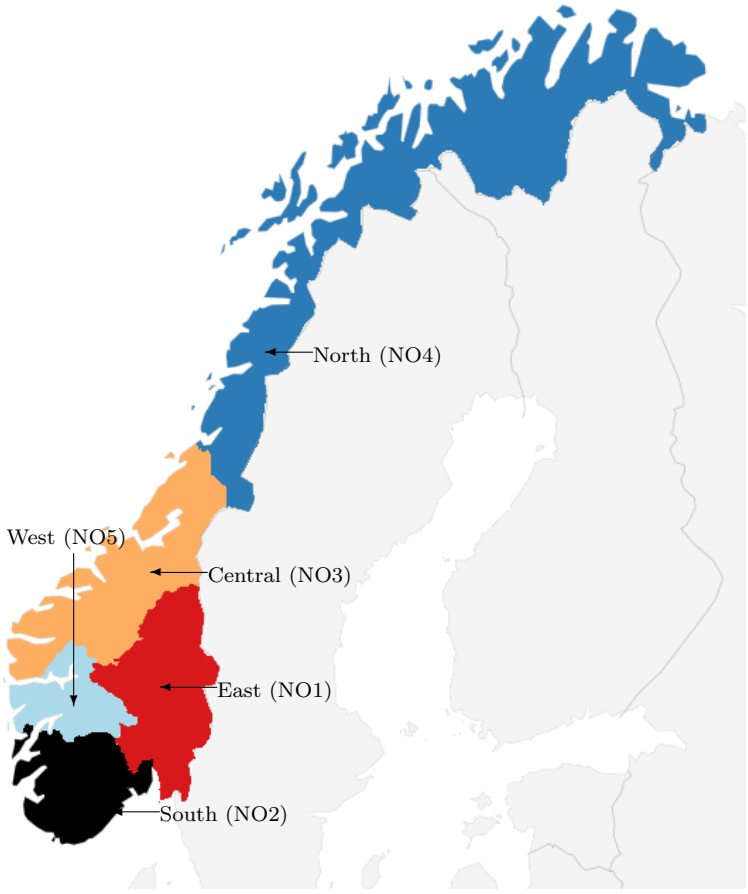
[6] Joskow and Tirole (2005), LaRiviere and Lyu (2022), and Lamp and Samano (2023) also study problems related to interconnection in electricity markets.

[7] See regjeringen.no and eea.europa.eu.

[8] Zonal pricing is also used in other countries. See for instance Eicke and Schittekatte (2022).

[9] See SSB (2025); and NVE–RME: Norway's smart-meter journey.

Figure 1: Norway's electricity bidding zones (NO1–NO5).



Notes: The electricity market in Norway is divided in five interconnected bidding zones. Zones 3 and 4 were largely shielded from the energy crisis. Zones 1, 2, and 5 were impacted by electricity price increases.

to the Netherlands (700 MW), NordLink to Germany (1,400 MW), and the North Sea Link to the United Kingdom (1,400 MW).[10] All these links provide more than 5.2 GW of direct interconnection capacity to foreign markets in the southern zones, while NO3 and NO4 have no comparable direct international connections.

Those differences in interconnection capacity played a key role in the divergence of wholesale electricity prices across regions. While prices in NO3 and NO4 often remained a fraction of those observed in the South, at times close to zero, Southern Norway experienced historically high prices during the peak of the crisis in 2022. According to Statistics Norway, the

[10] See www.statnett.no.

average household electricity price (excluding taxes and grid rent) reached 214 øre/kWh in the third quarter of 2022, almost five times higher than the corresponding five-year average of about 44 øre/kWh, representing the highest price recorded since the quarterly price statistics began in 1998.[11] Even after accounting for the government's electricity support scheme, introduced in December 2021, the total price paid by households averaged 141.5 øre/kWh, remaining about 30 percent above historical norms.

Importantly, these figures represent national averages and therefore mask substantial regional disparities. As emphasized by Statistics Norway, there has been a large and persistent price difference between Northern and Southern Norway, with households in the South exposed to much higher electricity prices throughout the crisis period.[12] As a result, Norwegian households mainly in the south have experienced high electricity bills.

In response to the financial burden imposed by these high price levels, the government introduced an electricity subsidy program in December 2021 that compensated households for the share of the electricity price above a fixed threshold (initially NOK 0.70 / kWh), later expanded in generosity and duration. As these policy adjustments unfolded in step with market conditions, the government's support evolved in discrete stages, initial 55% coverage above the threshold (December 2021), expansion to 80% and then 90%, a shift from monthly to hourly settlement in September 2023, and higher thresholds in 2024–2025. This sequence is summarized in Table 1.

Table 1: Electricity subsidy phases

Phase	**Period**	**Coverage and Threshold**	**Basis**
1	Dec 2021	55% coverage above 0.70 NOK/kWh	Monthly avg. price
2	Jan–Aug 2022	80% coverage above 0.70 NOK/kWh	Monthly avg. price
3	Sep 2022–Mar 2023	90% coverage above 0.70 NOK/kWh	Monthly avg. price
4	Apr–May 2023	80% coverage above 0.70 NOK/kWh	Monthly avg. price
5	Jun–Aug 2023	90% coverage above 0.70 NOK/kWh	Monthly avg. price
6	Sep–Dec 2023	90% coverage above 0.70 NOK/kWh	Hourly spot price
7	Jan–Dec 2024	90% coverage above 0.73 NOK/kWh	Hourly spot price

Notes: The subsidy threshold was constant throughout most of the sample period and was only slightly increased in January 2024. The calculation basis shifted from monthly average electricity prices (Phases 1–5) to hourly spot prices (Phases 6–7).

The introduction of electricity subsidies was a significant policy change in Norway's energy

[11] 1 NOK = 100 øre.
[12] See SSB (2023).

market. Although the program was initially intended as a temporary relief measure, it quickly became one of the largest fiscal responses to the energy crisis in Europe. The program's rapid expansion reflected the severity of regional disparities and the government's efforts to protect households from unprecedented price shocks.

However, the program also raised important questions about efficiency and distribution. By compensating consumers above a fixed threshold, the program reduced incentives to reduce demand during periods of scarcity. At the same time, since only certain regions experienced extremely high prices, the subsidies reinforced the two-tier market structure. Understanding how households responded to these subsidies, whether by reducing, maintaining, or increasing their electricity consumption, provides crucial insights into the interplay between market prices and government intervention.

3 Data

Our empirical strategy is based on two complementary datasets. We begin with an hourly bidding-zone panel that we use to describe stylized facts about the policy and to estimate price elasticities in a framework that explicitly reflects the subsidy schedule. We then introduce monthly administrative household microdata, which provide the granularity needed to estimate changes on electricity consumption and to understand the heterogeneity of such changes. Due to confidentiality protections and the aggregation structure of the metering records, the household panel cannot be linked to the hourly zonal series at the household or meter level.

3.1 High-frequency data

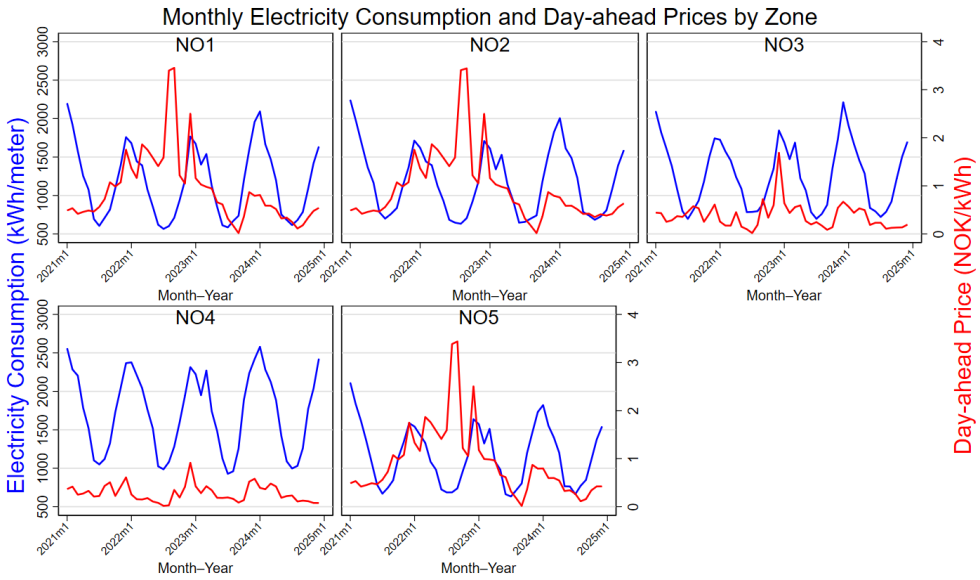
The high-frequency electricity consumption records are collected from Elhub and cover Norway's five bidding zones (NO1–NO5) from January 2021 to December 2024. We observe the total volume of electricity consumed and the number of metering points reporting consumption in each zone at the hourly level. These hourly consumption data are then merged with hourly day-ahead wholesale electricity prices sourced from Nord Pool, the Nordic electricity exchange operating the common power market for Norway and neighboring countries.[13]

[13]See nordpoolgroup.com. In the Nordic electricity market, the main wholesale electricity price is determined in the day-ahead auction operated by Nord Pool. The day-ahead price constitutes the spot price in Norway. After the day-ahead market clears, an intraday market allows for limited adjustments, but these trades involve relatively small volumes and do not replace the day-ahead price as the reference price for retail

These prices are set one day in advance through market clearing and vary by bidding zone, reflecting both local hydrological conditions and interconnection constraints.

Figure 2 illustrates the evolution of monthly average electricity consumption and day-ahead prices across the five zones. Consumption (blue) follows a strong seasonal pattern, peaking in winter months and declining during summer, consistent with heating demand. Prices (red) show much greater regional variation: zones NO1, NO2, and NO5 in the South experienced pronounced price spikes during the 2021–2023 crisis period, whereas NO3 (Central) and NO4 (North) remained largely insulated, with wholesale prices staying relatively low and stable.

Figure 2: Monthly electricity consumption and average day-ahead prices by zone



Notes: The left-hand y-axis shows the average monthly electricity consumption per meter in each bidding zone (kWh), while the right-hand y-axis displays the average hourly day-ahead price per month (NOK/kWh). The graph presents these two variables over time for each bidding zone (NO1–NO5).

It is also important to control for other external factors that may influence electricity consumption independently of price. For example, temperature plays a key role as people naturally use more electricity when it is cold. In addition, to account for market conditions when estimating the price elasticity, we combine data from several independent sources. Specifically, we obtain hourly wind-generation data for each Norwegian bidding zone from

contracts.

Elhub and national reservoir filling levels from the ENTSO-E Transparency Platform to capture variation in renewable generation and hydrological conditions.[14] We also obtain daily European natural gas prices from Bloomberg's EGSI NCED Index.[15] Given the close link between natural gas and electricity prices in European markets, this index provides an external benchmark for more general market conditions.

The final dataset consists of hourly observations for five bidding zones (NO1–NO5) from January 2021 to December 2024, yielding a balanced panel of 174,697 zone-hour observations. For each bidding zone, total hourly electricity consumption is divided by the number of household-level metering points in that zone to obtain average hourly consumption per household meter. All price series are expressed in NOK/kWh.

Table 2 summarizes descriptive statistics of key variables by zones. The average hourly electricity consumption per meter exhibits a range from approximately 1.6 kWh in Zones NO1, NO2, and NO5 to 2.4 kWh in Zone NO4, thereby indicating a higher baseline demand in the northern regions. Electricity prices also vary substantially across zones. For instance, Zones NO1, NO2, and NO5 face average prices close to 0.9 NOK/kWh, while Zones NO3 and NO4 exhibit considerably lower average prices. Thus, the observed price variations can be attributed to the presence of regional differences in market conditions. Temperature variation is also substantial, with mean values ranging from 3.9°C in Zone NO4 to 8.6°C in Zone NO2. Consistent with these patterns, heating degree days (HDD) are highest in Zones NO3 and NO4, while cooling degree days (CDD) remain low across all zones, thereby indicating that heating demand is the predominant factor influencing electricity consumption in Norway during the examined sample. The natural gas price index (NOK) is identical across Norwegian bidding zones, as it reflects a common European benchmark rather than a local market price. The original series is reported in EUR/MWh and has been converted to NOK/MWh. To ensure consistency with our hourly analysis, we further expand the daily observations to an hourly frequency by assigning each day's value to all hours within that day. The wind generation variable shows substantial and volatile hourly output (means from about 87 to 662 MWh and large standard deviations), consistent with meaningful installed wind capacity and strong weather-driven intermittency across zones NO1–NO4. In contrast, NO5 has a lower number of observations and an hourly mean close to zero (max 0.006 MWh), indicating that there is less wind production recorded within the NO5 bidding area.

[14] See entsoe.eu.

[15] Retrieved from Bloomberg: Powernext Gas Spot Zeebrugge ZTP European Gas Spot Index (EGSIZPAD Index, source PEGAS, frequency daily, unit EUR/MWh).

In fact this area remains dominated by hydro and has very limited installed wind-generation capacity.[16]

3.2 Household-level microdata

To understand electricity consumption patterns, we use monthly administrative household-level microdata which contain anonymized information on individual households' electricity consumption, combined with detailed socioeconomic and dwelling characteristics. These data are obtained from administrative registers and are accessed exclusively within a secure research environment.[17] The high-frequency data and the microdata differ in both temporal resolution and aggregation. Elhub data provide hourly electricity consumption aggregated at the bidding-zone level, whereas the household-level data from Microdata.no are derived from Elhub's metering records but are available at the household level and at a monthly frequency.

We restrict the analysis to primary residential dwelling types, namely detached houses, semi-detached houses, and low-rise multi-unit dwellings. When households are associated with multiple metering points, electricity consumption is aggregated to the household level by summing consumption across all meters registered to the same household within the same municipality.[18] The household-level microdata are used exclusively to analyze electricity consumption patterns and their heterogeneity across income groups, housing characteristics, and household composition. Table 3 reports descriptive statistics for all household-level variables.

Because electricity demand in Norway is highly sensitive to temperature, weather conditions constitute an important confounding factor. We therefore explicitly control for weather variation when estimating household-level consumption patterns. Hourly temperature data are obtained from the Norwegian Meteorological Institute and aggregated to the monthly level to match the frequency of the household electricity consumption. Given the lack of more granular spatial coverage, we assign weather conditions at the bidding-zone level using one representative city per zone: Oslo (NO1), Stavanger (NO2), Trondheim (NO3), Tromsø (NO4), and Bergen (NO5). From these temperature series, we construct heating and cooling degree days (HDD and CDD, respectively), defined relative to a 17°C baseline, which are

[16] Norwegian Ministry of Energy (energifaktanorge.no) and ssb.no.

[17] See https://www.microdata.no/.

[18] Residential electricity consumption also reflects slow-moving household and dwelling characteristics, implying that adjustment to energy cost shocks may be gradual rather than immediate (Brounen et al., 2012).

Table 2: Descriptive statistics by zone: hourly observations covering January 2021 through December 2024.

Variable	Mean	SD	Min	Max	N
Zone NO1					
Consumption (kWh)	1.602	0.690	0.527	3.889	34939
Electricity price (NOK)	0.923	0.853	0.000	7.767	34939
Temperature (°C)	7.432	8.651	-22.900	31.000	34939
HDD	10.049	7.927	0.000	39.900	34939
CDD	0.481	1.528	0.000	14.000	34939
Natural gas price index (NOK)	0.554	0.380	0.159	2.509	34939
Wind generation (MWh)	87.247	79.665	0.000	360.576	30530
Zone NO2					
Consumption (kWh)	1.620	0.645	0.581	3.795	34940
Electricity price (NOK)	0.944	0.843	0.000	8.983	34940
Temperature (°C)	8.625	7.681	-14.200	29.700	34940
HDD	8.813	7.034	0.000	31.200	34940
CDD	0.438	1.346	0.000	12.700	34940
Natural gas price index (NOK)	0.554	0.380	0.159	2.509	34940
Wind generation (MWh)	479.270	385.025	0.000	1371.671	30530
Zone NO3					
Consumption (kWh)	1.736	0.636	0.622	3.717	34940
Electricity price (NOK)	0.371	0.413	0.000	5.728	34940
Temperature (°C)	6.865	7.727	-17.500	30.400	34940
HDD	10.396	7.286	0.000	34.500	34940
CDD	0.261	1.090	0.000	13.400	34940
Natural gas price index (NOK)	0.554	0.380	0.159	2.509	34940
Wind generation (MWh)	661.693	515.687	0.000	1916.776	30530
Zone NO4					
Consumption (kWh)	2.365	0.775	0.780	4.376	34939
Electricity price (NOK)	0.280	0.292	0.000	4.901	34939
Temperature (°C)	3.916	7.277	-13.900	30.000	34934
HDD	13.218	6.976	0.000	30.900	34939
CDD	0.136	0.844	0.000	13.000	34939
Natural gas price index (NOK)	0.554	0.380	0.159	2.509	34939
Wind generation (MWh)	292.810	202.557	0.041	1009.746	30528
Zone NO5					
Consumption (kWh)	1.592	0.591	0.560	3.624	34939
Electricity price (NOK)	0.920	0.846	0.000	7.767	34939
Temperature (°C)	7.819	6.601	-11.400	29.600	34939
HDD	9.420	6.158	0.000	28.400	34939
CDD	0.240	1.065	0.000	12.600	34939
Natural gas price index (NOK)	0.554	0.380	0.159	2.509	34939
Wind generation (MWh)	0.001	0.001	0.000	0.006	26905
Weekly Data					
Reservoir filling (%)	58.04	16.97	20.94	85.99	156

Notes: Statistics use all available observations per variable within each zone; missing values are excluded variable-wise. The natural gas price index is converted from EUR to NOK using year-specific exchange rates. Wind generation is reported in MWh (kWh/1000).

Table 3: Descriptive statistics: household-level monthly data

	Mean	Std. Dev.	N	p1	p50	p99
Income (NOK)	802,749	482,075	53,181,832	191,478	719,287	3,036,267
Cons (KWh)	1502.1	1022.4	53,750,088	0	1273.1	5190.5
Log(Cons)	7.107	0.712	52,783,891	4.928	7.165	8.559
Usable floor area (m^2)	158.38	66.14	53,027,169	41	151	359
Household size (persons)	2.40	1.26	53,750,088	1	2	6
Building vintage	1971.6	32.9	52,551,434	1837	1977	2019
Number of rooms	4.62	1.58	49,675,929	1	4	9

Notes: The table reports descriptive statistics for household-level monthly microdata merged with electricity consumption records. All variables are observed at the household–month level. The sample contains approximately 1.5 million unique households (household–month observations aggregated over the sample period; the exact number varies slightly across variables due to missing values and sample restrictions). Percentiles (p1, p50, and p99) are computed over the full estimation sample covering January 2021 to December 2023. Data for 2024 are excluded due to capacity constraints in the microdata.no research environment. Household income is measured annually and fixed at its 2021 level for each household.

included as controls in all the regressions.[19]

4 The Consequences of the Structure of the Subsidy

The electricity subsidy scheme was implemented in phases, each characterized by different coverage rates and price thresholds. In general, the subsidy *per kWh* at time t can be described as

$$\text{subsidy}_t = K_t \times \max\{0, p_t - p^*\},$$

where p_t is the electricity price at time t and p^* is the price threshold above which the subsidy is binding. The price to compute the subsidy, p_t, has been defined in slightly different ways at different phases of the policy as explained below, but for expositional purposes we will treat it as the spot price of electricity. The difference $p_t - p^*$, when positive, determines the subsidy spread. K_t is the coverage rate and is a constant between 0 and 1 that represents the amount of the spread that is reimbursed to consumers. As explained below, since December 2021, p^* has taken the values 0.70 and 0.73 NOK / kWh at different points in time, and

[19] HDD and CDD are constructed from daily average temperature T_{zt} in zone z and period t relative to a baseline of 17°C. Specifically, $\text{HDD}_{zt} = \max\{0, 17 - T_{zt}\}$ and $\text{CDD}_{zt} = \max\{0, T_{zt} - 17\}$. A 17°C base is commonly used in Nordic applications; see, e.g., Cox et al. (2015) for Denmark and Hilliaho et al. (2015) for Finland.

K_t has been set at 0.55, 0.80, and 0.90. In other words, the subsidy covers a percentage of the excess price above the threshold. For example, suppose the price threshold is $p^* = 0.70$ NOK/kWh, or 70 øre/kWh, and the reimbursement rate is $K_t = 0.90$. The consumer pays the full price up to the threshold but only 10% of any amount exceeding it. Thus, if $p_t = 70$ øre/kWh, the consumer pays 70 øre/kWh. If $p_t = 71$ øre/kWh, the net price is $70 + (1 - 0.90) \times (71 - 70) = 70.1$ øre/kWh.

We revisit Table 1, which summarizes the subsidy design across the seven phases. Up to Phase 5 (January 2022–August 2023), the subsidy was calculated using the monthly average electricity price. Beginning with Phase 6 (September 2023), the calculation shifted to the hourly spot price, which continued into Phase 7 (January–December 2024). Yet, the coverage rate increased from 55 percent in December 2021 (Phase 1) to 80 percent in early 2022 (Phase 2), and further to 90 percent between September 2022 and March 2023 (Phase 3). Coverage was temporarily reduced to 80 percent during April–May 2023 (Phase 4), before returning to 90 percent in June–December 2023 (Phases 5-6). In January 2024 (Phase 7), coverage remained at 90 percent, though the price threshold was slightly raised from 0.70 NOK/kWh to 0.73 NOK/kWh.

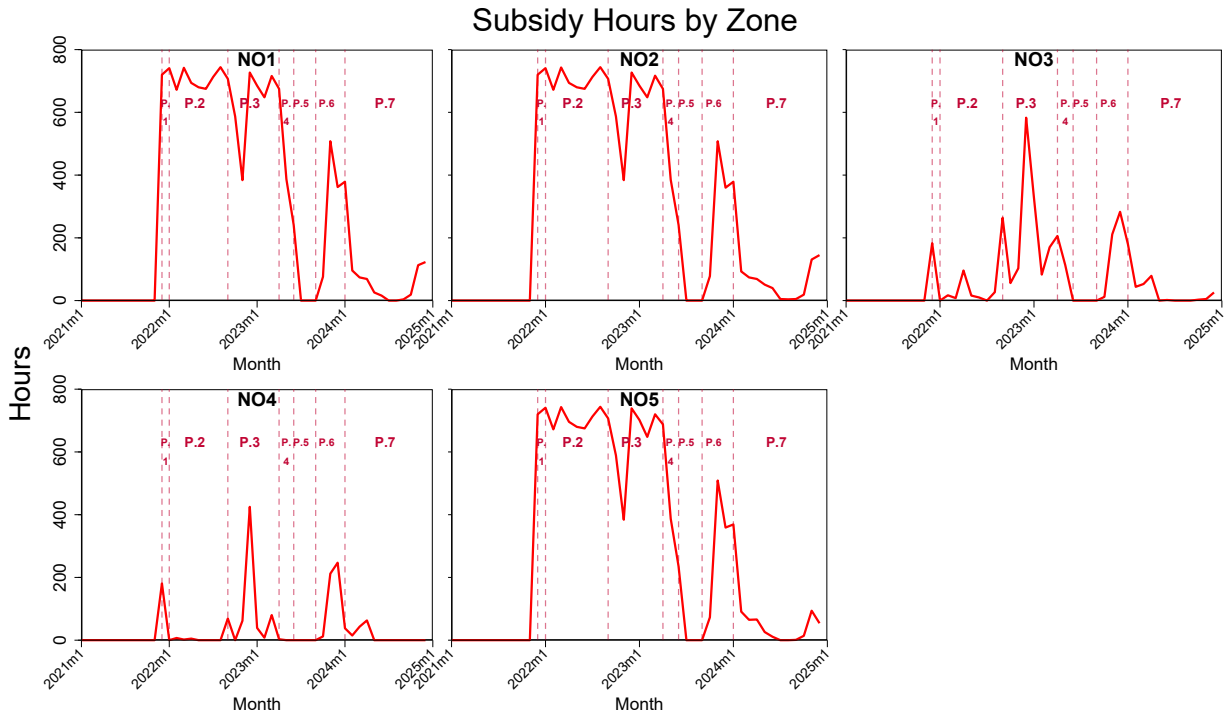
4.1 Three facts about the electricity subsidies in Norway

To illustrate how these rules translated into observable market outcomes, we present three descriptive figures.

1. The subsidy was active almost all the time immediately after the subsidy started. Figure 3 shows the frequency of subsidy hours per month in each zone, capturing the intensity of the scheme across time and zone. Subsidy activity increased sharply from late 2021 onward, consistent with the introduction of the scheme, and remained concentrated in Zones 1, 2, and 5. In contrast, Zones 3 and 4 experienced limited subsidy exposure, reflecting lower average prices. The figure also illustrates the drop in subsidy hours during 2023 before a renewed increase in early 2024, in line with the price patterns observed in Figure 2.

2. Subsidy amounts were negligible after mid-2023. Figure 4 shows the average monthly subsidy by zone. Subsidy expenditures track both consumption levels and wholesale price volatility, with pronounced peaks during the winter of 2022–2023 when subsidies exceeded 2,500 NOK per household on average in Zones 1, 2, and 5. Conversely, Zones 3

Figure 3: Frequency of subsidy activity



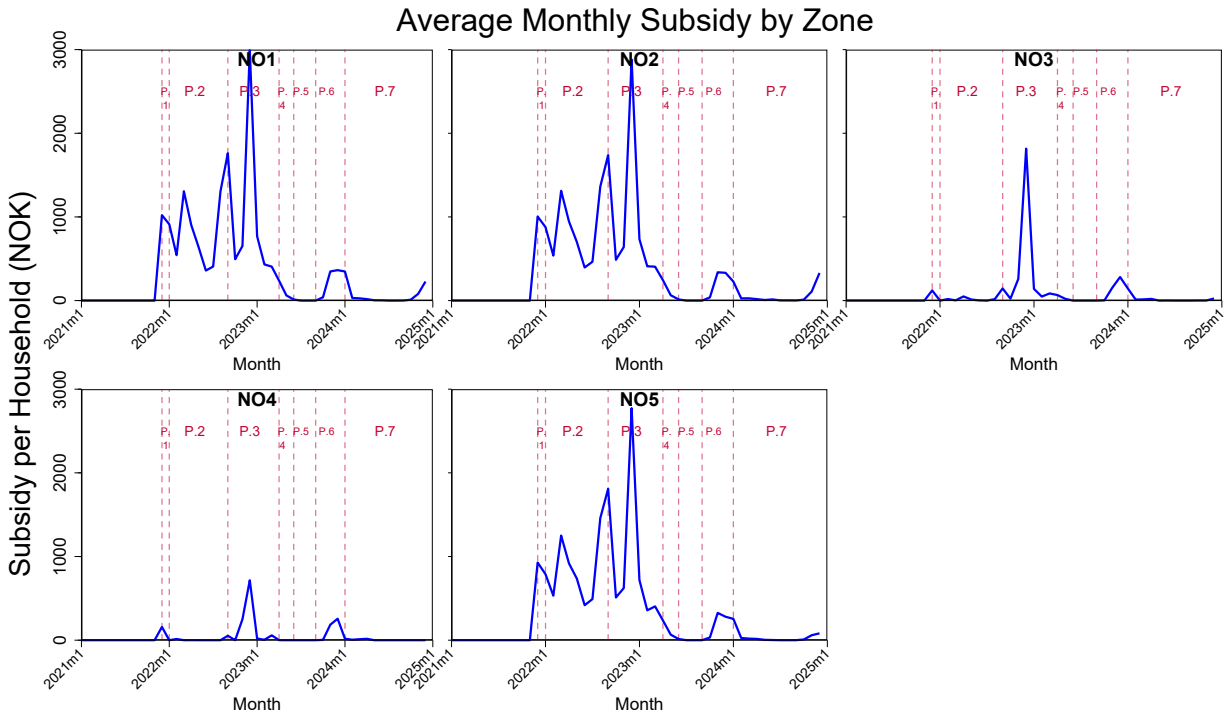
Notes: The figure shows the number of hours per month in which the subsidy was binding, by electricity zone. Zones NO1, NO2, and NO5 experienced frequent subsidy hours due to higher wholesale prices, while zones NO3 and NO4 show limited subsidy activity. Vertical lines reflect changes in the subsidy design (see Table 1).

and 4, which experienced lower prices, received significantly less compensation. The heterogeneity in the subsidy compensation highlights the variation in treatment intensity across zones. Despite that hundreds of hours per month were eligible for the subsidy, after mid-2023 (Figure 3), the monthly average subsidy amounts were largely negligible.

3. Divergence of net and spot prices occurred in only two Phases. Figure 5 shows the relationship between day-ahead market prices and the effective consumer prices net of subsidy. The substantial gap between the two series in Phases 2 and 3 highlights the extent to which subsidies shielded households from significant price increases across zones, particularly during 2022 and the first half of 2023. This is particularly evident in Zones 1, 2, and 5, consistent with the previous two facts about the subsidy outcomes.

Relative to the pre-subsidy average, calculated over 11 months before the policy, average

Figure 4: Subsidy expenditures



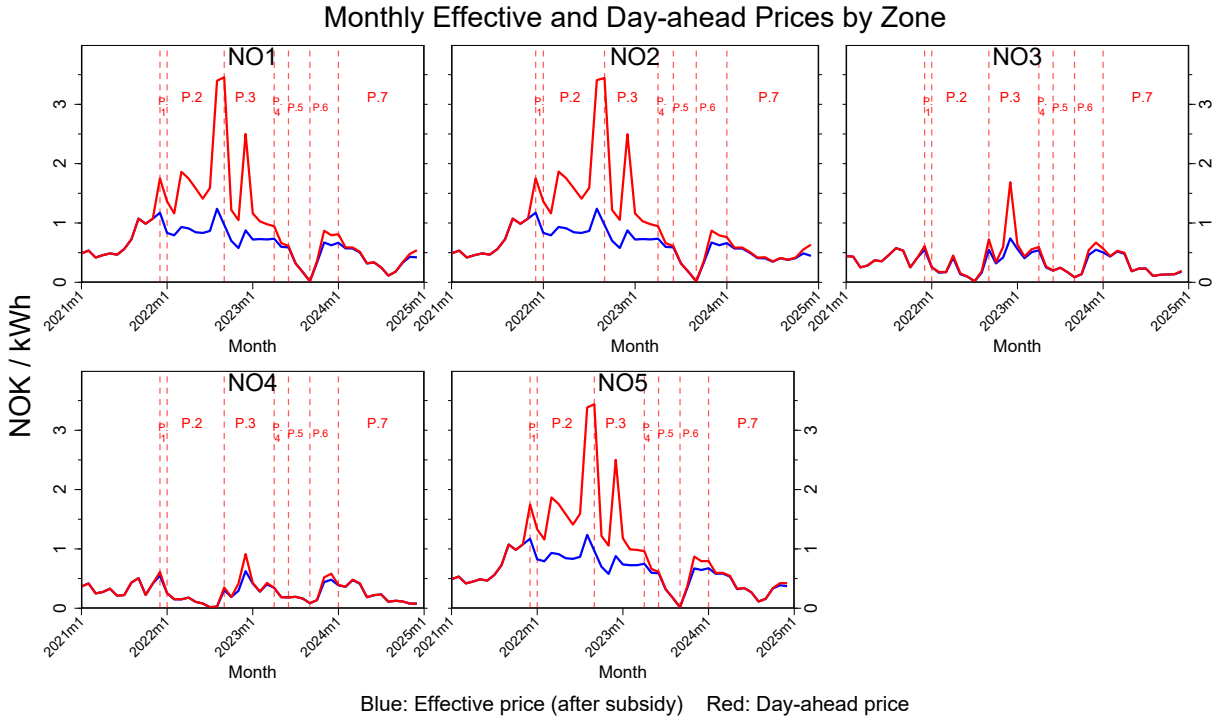
Notes: The figure shows average monthly subsidy compensation on electricity bills per household (NOK) across zones. Vertical dashed lines indicate changes in the subsidy design (see Table 1).

effective—or net—prices across bidding Zones 1, 2, and 5 increased by approximately 77.2% in Phase 1, 37.2% in Phase 2, and 14.6% in Phase 3. These changes are very similar when calculated separately for each of those zones. In Phase 4, average net prices were close to pre-policy levels, while in Phases 5–7 they fell below pre-policy averages.

4.2 A regression kink design

The definition of the subsidy gives place to a continuous price function with a change in slope (a kink). The slope of the relationship between the spot price and net price changes at the threshold p^*, but the level does not. We can use a regression kink design (RKD) (Card et al. (2015)) to exploit this change in slope. A simple representation of this situation is shown in Figure 6. There, the net price coincides with the spot price up to the threshold value along the 45-degree line. To the right of the threshold, the net price line has a slope of $1 - K_t$, thus a lower and different slope than to the left of the price threshold.

Figure 5: Day-ahead electricity prices and net prices under the subsidy



Notes: The net price is defined by the policy as $p^* + (1 - K_t)(p_t - p^*)$ whenever $p_t > p^*$, and equal to the hourly price otherwise. p^* is the price threshold, K_t is the fraction paid by the government, and p_t is the hourly price. Dashed vertical lines mark changes in the subsidy design (Jan 2022, Sep 2022, Apr 2023, Jun 2023, Sep 2023, Jan 2024).

We estimate the change in the slope of log consumption (Y_t) relative to the assignment variable (p_t) with the following model,

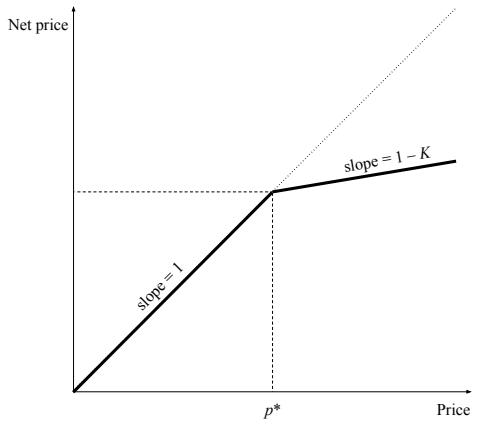
$$Y_t = \alpha + \tau_1(p_t - p^*) + \tau_2 \mathbb{1}(p_t > p^*) \times (p_t - p^*) + \mathbf{X}_t'\beta + \boldsymbol{\delta}_{\cdot} + \epsilon_t \tag{1}$$

where p_t is the running variable (hourly price). $\boldsymbol{\delta}_{\cdot}$ is a general representation for a variety of fixed effects and their combinations (by zone, hour of the day, and day of the week). $\mathbf{X}_t'$ is a vector of controls that include temperature, a natural gas price index, wind generation, and percentage reservoir fulfillment. We estimate this model using only the zones and phases where the subsidy is materially binding. This occurs in Zones 1, 2, and 5 (see section 2), and during Phases 2 and 3 (Figure 5). We elaborate on this requirement further below.

The causal effect is identified by the ratio of the change in consumption slope to the change in price slope:

$$\eta_{RKD} = \frac{\text{change in slope of consumption at threshold}}{\text{change in slope of price at threshold}} = -\frac{\tau_2}{K_t},$$

Figure 6: Regression kink design



Notes: The net price of electricity coincides with the spot price up to the threshold p^* (slope of 1). However, the price schedule changes sharply in slope exactly at the p^* threshold, to a slope of $1 - K$. We omit the time index to ease readability. Price data points can fall on either side of the p^* threshold yielding quantity demanded observations affected by one of the two different net price slope levels.

where K_t is the coverage given by the subsidy.

The obvious concern with the running variable is that it is endogenous: spot prices are determined by the intersection of supply and demand. The source of the bias is that high prices usually correlate with higher consumption (due to cold weather for instance). To address this concern, we added an extensive set of controls in the form of polynomials of temperature and other variables. This helps with identifying the residual consumption change driven by the price signal, distinct from the weather signal.[20].

The standard instrumental variable framework fails in this context for two primary reasons. First, the instrument (the activation of the subsidy) is a deterministic function of the endogenous running variable (the spot price). In electricity markets, high spot prices are perfectly correlated with unobserved structural demand shocks, such as extreme weather events. Consequently, the instrument violates the exclusion restriction unless the spot price is perfectly controlled for—but doing so linearly creates perfect collinearity, collapsing the first stage. Second, a linear instrumental variable model masks the structural break created by the policy, averaging out the sharp, non-linear variation in the marginal price schedule that is the actual source of exogenous variation.

[20] Another potential concern is that the density of the running variable might exhibit bunching around the threshold. If this were the case, it would invalidate the RKD estimator. However, an examination of the price density shows no evidence of bunching around the threshold or elsewhere. See Figure A.1 and further details in subsection A.1

Instead, the consumer's marginal price schedule exhibits a sharp change in slope exactly at the p^* threshold (from a slope of 1 to a slope of $1 - K$) as shown in Figure 6. By restricting the bandwidth to hours immediately surrounding the cutoff, we can rely on the standard RKD identifying assumption: the direct marginal effect of unobserved confounders (e.g., temperature) on consumption evolves smoothly through the 70 øre threshold. Under this assumption, any observed change in the slope of the conditional expectation of log consumption at the threshold is non-parametrically identified as the causal effect of the marginal price change. The choice of such bandwidth is explained in detail further below.

This framework recovers a Local Average Treatment Effect (LATE), in this specific case, the price elasticity of demand evaluated precisely at the high-price threshold. In our context, this isolates consumer responsiveness exactly when the grid is most constrained and the marginal costs of generation are at their peak.

To isolate this effect, it is necessary that other factors do not exhibit a structural break at this particular price level. Figures A.2 - A.5 in the Appendix show that temperature, wind speed, fulfillment reservoir levels, and natural gas prices, which are the covariates included in the regressions, do not have a change in slope around the price threshold in each of the two phases. In addition, we assume that unobserved determinants of consumption are smooth functions of the spot price. Therefore, any change in the slope of consumption with respect to the spot price at the threshold must be driven by the change in the slope of the price faced by the consumer.

Back to Equation 1, we expect τ_1 (the baseline slope) to be negative and τ_2 (change in slope) to be positive. τ_1 captures the relationship between consumption and spot prices below the threshold, where consumers pay the full price. As the price of electricity rises, households typically reduce consumption or at least, they do not increase it. The demand curve slopes downward. This full exposure should discourage consumption. τ_2 is the flattening effect coefficient on price above the kink. This coefficient captures the difference between the slope above the threshold and the slope below the threshold. Because the consumer is protected from price spikes above p^*, their demand should become less responsive to the spot price. The relationship becomes flatter (closer to zero). In other words, we are moving from a steep negative slope (e.g., -0.5) to a flat negative slope (e.g., -0.05).

The RKD estimator essentially asks: if the price signal gets weaker, does the consumption reaction also get weaker (positive change relative to trend)? Since the adverse effect of high prices on consumers was removed by the subsidy (the denominator), the reduction in usage

usually caused by high prices disappeared (the numerator). The ratio of these two movements recovers the underlying negative relationship between price and quantity.

Table 4 shows the estimation results of Equation 1 for Phase 2 (top part of the table) and for Phase 3 (bottom), which are periods of time where the policy was binding most of the hours. Subsequent phases (and Phase 1) do not have enough hours with a net price different from the spot price (as seen in Figure 5). This is crucial for identification of the RKD model. We emphasize that the sample uses Zones 1, 2, and 5 only since these are the only zones with a gap between the net and the spot price in both Phases 2 and 3. We show three different specifications that include different combinations of fixed effects. The basis for the calculation of the subsidy during these two phases is the monthly average price, however we use the hourly frequency data since otherwise we would have only a handful of observations during the 8 months of the Phase 2 and the 7 months of Phase 3. The coefficients have the expected signs and are strongly statistically significant in both phases, except for the coefficient of the demand slope (τ_1) in Phase 3, specifications (1) and (2). This has no effect on the elasticity calculations below since this coefficient does not enter the expression for the elasticity. The demand slope (τ_1) is fairly stable across the different specifications, while the coefficient on the "correction term" (τ_2) is slightly more sensitive to the addition of fixed effects in Phase 2 but fairly stable in Phase 3. Although we used the hourly data for the estimation, the number of observations is small because of the bandwidth chosen of 30 øre for Phase 2 and of 20 øre for Phase 3, around the threshold. Several robustness checks are shown in subsection A.3 in the Appendix regarding the sensitivity of the results to the choice of the bandwidth.

For each phase we also report the semi-elasticities η_{RKD} calculated as explained above in this subsection and the implied elasticities. The standard errors were computed using the Delta method. The semi-elasticity estimator from the regression kink design follows the formula:

$$\eta_{RKD} = \frac{\Delta\text{Slope of } \log(Y)}{\Delta\text{Slope of } p}.$$

Because the dependent variable is in logs ($\log Y$) and the independent variable is in levels (p), this ratio represents a semi-elasticity. To transform it into an elasticity, we multiply our RKD estimator by the price level at the point of estimation (the threshold p^*).

$$\epsilon_{RKD} = \text{Semi-Elasticity} \times p^* = \eta_{RKD} \times p^*.$$

The resulting elasticities are shown in the last row for each Phase with standard errors

Table 4: Estimates of the regression kink design

	Phase 2		
	(1)	(2)	(3)
Price centered (τ_1)	-0.357**	-0.419**	-0.308**
	(0.154)	(0.190)	(0.126)
Price above thres. (τ_2)	0.808***	1.248***	0.724***
	(0.245)	(0.298)	(0.195)
N	518	518	518
R^2-adj.	0.85	0.76	0.88
DV mean	0.30	0.30	0.30
Zone FE	No	Yes	Yes
Hour of the day FE	Yes	No	Yes
Day of week FE	Yes	No	Yes
Semi-elasticity and elasticity values			
$\eta = \frac{\Delta \text{ slope of } \log q}{\Delta \text{ slope of } p}$	-1.010^{***}	-1.560^{***}	-0.905^{***}
	(0.306)	(0.372)	(0.244)
$\eta \times p^*$	-0.707^{***}	-1.092^{***}	-0.634^{***}
	(0.215)	(0.261)	(0.171)
	Phase 3		
Price centered (τ_1)	-0.166	-0.198*	-0.168
	(0.111)	(0.111)	(0.108)
Price above thres. (τ_2)	0.601***	0.720***	0.600***
	(0.173)	(0.153)	(0.171)
N	1,581	1,581	1,581
R^2-adj.	0.60	0.53	0.60
DV mean	0.55	0.55	0.55
Zone FE	No	Yes	Yes
Hour of the day FE	Yes	No	Yes
Day of week FE	Yes	No	Yes
Semi-elasticity and elasticity values			
$\eta = \frac{\Delta \text{ slope of } \log q}{\Delta \text{ slope of } p}$	-0.668^{***}	-0.799^{***}	-0.666^{***}
	(0.192)	(0.170)	(0.189)
$\eta \times p^*$	-0.467^{***}	-0.560^{***}	-0.466^{***}
	(0.134)	(0.119)	(0.133)

Notes: For each of Phases 2 and 3, the table shows the results from Equation 1 using hourly data with a bandwidth of 0.30 NOK / kWh and 0.20 NOK / kWh, respectively. Data from Zones 1, 2, and 5 only. All models include controls, see main text for details. Standard errors for the implied semi-elasticities and elasticities were obtained using the Delta method. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

calculated using the Delta method. The demand elasticities range, in absolute value, from 0.6 to 1.1 in Phase 2 and from 0.5 to 0.6 in Phase 3, and should only be taken as the consequence of a local average treatment effect around the threshold defined by the policy. All the elasticities are strongly statistically significant across specifications and phases.[21]

Figure A.6 in the Appendix shows the binscatter of residuals of the RKD regression as a function of the centered spot price. This is a visualization that confirms the validity of the RKD approach to this setting. The y-axis is the unexplained variation in consumption after removing out the effects of the weather and all the fixed effects. In both Phases 2 and 3, we see a change in the slope of those residuals. To the left of the kink consumers face 100% of the spot price. As the price increases towards 70 øre, the marginal cost of running a heater or washing machine goes up steadily. Therefore, the slope of the dots is downward sloping. As we move from -0.20 to 0 on the x-axis, consumption drops. Above the price threshold, the consumer is only paying a fraction $(1 - K)$ of the spot price increase. The disutility of price to consumers is almost entirely neutralized by the government. Because the net price has stopped rising so aggressively, the downward trend in consumption abruptly stops.

A vast literature has estimated demand elasticities for electricity in Norway prior to the implementation of the subsidy.[22] For completeness, we also conduct our own estimation and report our results in Appendix B. We employ a log-log model with prices instrumented by wind output and natural gas prices. We report elasticities for each zone separately across three samples: before the subsidy was implemented, during the subsidy period using raw prices, and during the subsidy period using prices net of the subsidy. While not central to our study, we estimated these elasticities to demonstrate that a traditional instrumental variable approach yields highly inelastic values for the subsidy period. We find that when using the net price as the independent variable and instrumenting for price, the resulting elasticity estimates are roughly twice as large as those obtained using the raw price. This discrepancy arises because traditional methods fail to account for the specific structure of

[21] It is useful to place these values within the European context. Ahlvik et al. (2025) report an elasticity of -0.18 during the 2022 Energy Crisis in Finland when prices doubled. Although our estimated elasticities are at least three times larger, the underlying domestic policy in each study is very different but both point towards an economically important change in the responsiveness of consumers. Using data from Spain around the financial crisis of 2008, Romero-Jordán et al. (2016) estimate household elasticities around -0.15 to -0.25 due to an accumulated 63% increase in prices during that period. These elasticities are also an economically relevant increase in the consumers' responsiveness due to a price increase during a crisis.

[22] For instance, Hofmann and Lindberg (2019) estimate short-term elasticities between -0.011 and -0.075 for the Oslo region. Earlier aggregate studies such as Bye and Hansen (2008) and Johnsen (2001) document values between -0.02 and -0.05.

the subsidy policy: the change of the slope of the price schedule.

It is important to note that the threshold (0.70 NOK / kWh) and the prices above it are significantly higher than the levels to which consumers are accustomed. Specifically, during the period from January to November 2021–the 11 months preceding the policy–the average price in Zones 1, 2, and 5 was 0.66 NOK / kWh. During Phase 2, the average raw price increased to 1.77 NOK/kWh, 168% above its pre-policy level, while the average net-of-subsidy price was 1.05 NOK/kWh, representing an increase of 59%. During Phase 3, the corresponding raw and net prices were 1.63 and 0.82 NOK/kWh, respectively, representing increases of 147% and 24% relative to the pre-policy average.

5 Discussion and Broader Context

A natural expected outcome for electricity consumption levels consistent with the elasticity values found in the previous section is that quantities drop significantly as price increases. The dataset used in the previous section does not allow us to measure quantity changes at the household level because of its level of aggregation, instead we make use of the second dataset, the microdata from Statistics Norway (see subsection 3.2) to describe changes in consumption patterns at the household level and their correlations with households characteristics, notably income.

5.1 Descriptive analysis of consumption patterns

We examine how electricity consumption changed during the time of the subsidy relative to the pre-subsidy period by exploiting regional variation in exposure to the European energy crisis with an event-study regression. Zones that are largely independent of electricity flows with the rest of Europe serve as the control group, while southern zones that are physically interconnected with other European electricity markets constitute the treated group as in Brolinson et al. (2026). This analysis is not meant to capture a causal effect of the subsidy on consumption since the model below does not disentangle the effect of prices increases from the effect of the subsidy alone. However, this analysis describes how consumption patterns change around the beginning of the policy period as a consequence of the net price increase. The main motivation to expand the analysis from the previous section is that it is an empirical question whether the changes in the net price—a combined effect of the supply shock from continental Europe and the policy in Norway—had a long-lasting effect on demand and the

magnitude of that effect over time.

To address that question, we construct a series of time dummies indicating the number of months before and after the subsidy or treatment. These are interacted with the treatment group indicator to estimate the monthly impact relative to the baseline month (November 2021), leading to the following model:

$$\log Y_{izt} = \sum_{\substack{k \neq -1 \\ k=-M_0}}^{M} \beta_k \big(\text{Treated}_z \times \mathbb{1}\{t = k\}\big) + \mathbf{X}_{zt}\boldsymbol{\beta} + \alpha_z + \gamma_m + \varepsilon_{izt}, \tag{2}$$

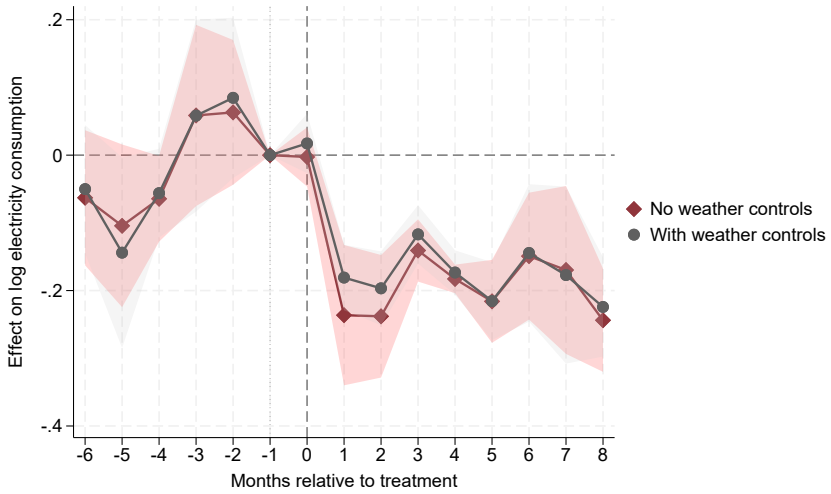
where $\log Y_{izt}$ denotes the log electricity consumption of household i in zone z and month t. The function $\mathbb{1}\{t = k\}$ is an event-time indicator equal to one if month t is k months relative to the treatment onset, with November 2021 ($k = -1$) omitted as the baseline period. The lower and upper bounds of the months considered relative to the event are $-M_0$ and M, respectively. Treated_z is an indicator equal to one for treated zones (NO1, NO2, and NO5, see Figure 1). $\mathbf{X}_{zt}$ contains weather-related control variables (temperature, temperature squared, HDD, and CDD). Zone fixed effects are denoted by α_z, and γ_m denotes calendar-month fixed effects that control for seasonal consumption patterns common across years. The error term ε_{izt} captures idiosyncratic shocks at the household-zone-time level. Standard errors are clustered at the zone level.

In Figure 7, we present the results from the specification in Equation 2. The vertical line at period 0 marks the first month of treatment (December 2021). This event study plot shows the treatment dynamics within this period at the monthly level, with and without controls. The results of the two specifications give practically the same results.

Before December 2021, the treatment month, the event-study coefficients are not statistically different from zero, providing no evidence of differential pre-trends between treated and control zones and supporting the parallel trends assumption. Following the onset of the price spike, the estimates turn sharply negative from January 2022 onward and are statistically significant across specifications. The implied demand reductions range from approximately 15 to 22 percent relative to the control zones, with the largest declines occurring in the first months after treatment. While the magnitude of the effect partially attenuates over time, consumption remains persistently lower for several months.[23]

[23]Extending the event-study window over a longer period is constrained by the secure Microdata.no environment as of August 2026. The specification requires constructing a separate set of relative-time indicators and their interactions with the treatment indicator, and extending these through the full sample exceeds the dataset-size limits currently supported by the platform.

Figure 7: Event Study: Dynamic Treatment Effects



Notes: Coefficients are normalized relative to month −1, the month immediately preceding treatment, which is omitted as the reference period. Shaded areas represent 95% confidence intervals based on clustered standard errors at the zone level. Estimates from data up to August 2022.

In Appendix C we present the results from a differences-in-differences regression model that extends the analysis shown in Equation 2. We consider the same outcome variable and the same time event to define the pre- and post-periods. We also keep the same definitions of the treated and control groups.[24] We find a 7–9% consumption drop as a result of the net price increase due to the crisis shock and despite generous government compensation relative to both the control group and the pre-crisis and pre-subsidy period. This is not a causal analysis of the subsidy on consumption, but rather a descriptive analysis to quantify the changes in consumption due to the net price increase during this period of time.

Whether we rely on elasticities from the existing literature or our own elasticities in Appendix Table B.2, electricity demand in Norway appears highly inelastic. Such values, however, fail to predict the relative drop in demand following a 37% net price increase. If we instead use the most conservative value of the elasticities obtained from the regression kink design in Table 4 (−0.63 in Phase 2), a 37% increase in net price implies a reduction in quantity demanded of 23.3%, a value that is much closer in magnitude to a raw comparison

[24]Following Brolinson et al. (2026), we examine the parallel-trends assumption by re-estimating the preferred specification (column (5) from Table C.1) using only the pre-shock period, January–August 2021. We interact the high-price-zone indicator with a linear monthly time trend. The estimated coefficient on this interaction is −0.01 (s.e. 0.019; p-value 0.52) and is statistically indistinguishable from zero. We therefore find no evidence that electricity consumption in the high- and low-price bidding zones followed different linear trends before the electricity-price shock.

of the consumption levels in the subsidy period and the pre-subsidy period.[25]

5.2 Income and other heterogeneous effects

To examine heterogeneity in consumption effects during the policy period, households are classified into discrete groups based on income, dwelling characteristics, and household composition. Then we use a differences-in-differences framework with the same control and treatment groups structure as in the event study above but instead of having multiple coefficients for each month, we only keep one coefficient for the months after the policy was implemented (the "Post" period). To examine heterogeneity by household income, we further interact the Treated × Post term with an indicator for each income group. We omit the lowest income group to avoid multicollinearity. In all specifications, the lowest category within each dimension serves as the omitted reference group.

Household income is grouped into five categories based on the empirical distribution of annual household income. Specifically, we construct income quintiles such that each group contains approximately 20 percent of households. The resulting income thresholds range from households earning below NOK 407,635 in the lowest quintile to households earning NOK 1,096,594 or more in the highest quintile. Mean income increases monotonically across groups, from NOK 298,727 in the lowest quintile to NOK 1,531,810 in the highest quintile.

Dwelling size is measured by total usable floor area and categorized into standard size intervals, which are subsequently aggregated into five groups reflecting increasing dwelling size. Building age is defined by year of original construction without including renovations and grouped into five categories corresponding to distinct building vintages. These vintages capture differences in building regulations, insulation standards, and heating technologies over time, with the oldest buildings serving as the baseline category. Housing complexity is proxied by the number of rooms and grouped into five categories reflecting increasing dwelling size and layout complexity, dwellings with two rooms or fewer form the reference group.

Household composition is measured by the number of registered household members. Households are classified into five categories ranging from single-person households to households with five or more members, with single-person households serving as the reference category.

[25] For example, with an elasticity of -0.08 (Hofmann and Lindberg (2019)), $\Delta\%q = -0.08 \times 37\% = -3\%$, which is about five times smaller than the drop in electricity demand in treated zones. Specifically, the average per-household hourly electricity consumption in the treated group dropped from 1.70 kWh in the 11-months before the policy started to 1.45 kWh during Phase 2, which represents a decrease of 14.8%.

Table 5: Heterogeneous treatment effects by income and household characteristics

	Group				
	2	3	4	5	N
Income group	-0.011^{***}	-0.017^{***}	-0.0204^{***}	-0.017^{***}	52,244,018
Used space (bruksareal)	-0.0017	-0.0104	-0.0208^{***}	-0.0249^{**}	52,097,156
Building year (byggeår)	0.0003	0.0053^{***}	0.0201^{***}	0.0275^{***}	51,642,742
Number of rooms	-0.0138^{***}	-0.0162^{***}	-0.0195^{***}	-0.0265^{***}	48,830,338
Household size (persons)	-0.0119^{***}	-0.0153^{***}	-0.0148^{***}	-0.0163^{***}	52,783,891
Fixed effects	Household, Month, Year, Zone				

Notes: The table reports the heterogeneity effects from regressions interacting post-treatment status, treatment group, and household characteristics. The dependent variable is log electricity consumption. Group 1 is the omitted reference group in each dimension (lowest income, smallest dwelling, oldest buildings, fewest rooms, and single-person households). Coefficients represent differences in treatment effects relative to the baseline category within the treated group. See Appendix Table C.6 for the full set of results and Table C.7 for the construction of household characteristic groups. Statistical significance is based on clustered standard errors at the zone level, not shown. $^{***}p < 0.01$, $^{**}p < 0.05$.

Table 5 reports the estimates obtained from separate regressions, each interacting treated households, the post-treatment period, and one of the household characteristics group indicators, while Table C.6 in the Appendix reports the entire set of results. The estimated coefficients capture differential effects relative to the baseline category, to the pre-period, and to the control group. The coefficients decrease systematically with dwelling size and number of rooms, and increase for newer buildings, consistent with differences in insulation quality and heating efficiency. In addition, we find strong heterogeneity across income groups. Relative to the lowest income quintile, households in the upper-middle income group reduce consumption by an additional 2 percentage points, and households in the highest income group by about 1.7 percentage points. This suggests that higher-income households have greater flexibility in adjusting electricity usage in response to price increases.[26]

[26] Garnache et al. (2025) find that Norwegian households with electric vehicles, whose ownership positively correlates with higher incomes, exhibit strong load-shifting during critical-peak pricing but achieve similar overall demand reductions across all income groups. The contrast with our findings suggests that income-driven response heterogeneity depends heavily on available technology and the specific nature and duration of the price shock.

5.3 The economic policy context

Finally, we compare the subsidy against an alternative fixed-price policy. As recently as October 2025, the government introduced a fixed-price option that co-exists with the subsidy policy. Consumers can choose whether to remain in the variable price regime with its corresponding subsidy or to switch to the fixed-price regime, the "Norgespris".[27]

The most important distinction between the subsidy and the fixed-price regime lies in the preservation of the marginal price signal. Under the fixed-price policy, the connection between the wholesale market price and what consumers pay is completely absent, similarly to the case in North American electricity markets. Because the consumer's marginal cost of consumption is invariant to spot market fluctuations, the marginal price signal is entirely lost. In contrast, the subsidy scheme preserves at least a portion of the marginal price signal.

This difference is crucial if similar conditions to those in 2022 repeat in the near future. During the crisis, the electricity market was frequently operating on the steep, highly inelastic portion of the aggregate supply curve. Under a fixed-price policy, households would have had no incentive to reduce their consumption during peak hours, forcing the system to clear at extremely high marginal generation costs. By preserving at least some of the marginal price signal, the subsidy incentivized at least some demand reduction during those hours. Precisely because of this demand response, a fixed-price policy would have ultimately resulted in substantially higher total system costs and their corresponding government expenditures compared to the subsidy regime.

6 Conclusion

We analyzed demand responsiveness in a substantial subsidy program designed to shield consumers from high electricity prices. Using a regression kink design, we assessed household behavior around the pre-established price threshold where the subsidy activates. We find a statistically significant coefficient regarding the change in the slope of the demand curve above the threshold; this quantifies and confirms the impact of the subsidy structure on demand curvature. This approach allows us to estimate a demand elasticity locally around the policy's price threshold. We find a relatively elastic demand curve at this point, suggesting

[27] The Norgespris is a fixed-price option for household electricity consumption, allowing consumers to pay a predetermined price of 0.40 NOK/kWh (excluding VAT) instead of being fully exposed to spot market prices. The scheme was announced in response to prolonged price volatility following the energy crisis and is intended as an alternative to direct subsidy-based support. See regjeringen.no for details.

that a standard constant-elasticity model would have failed to capture consumer behavior in this specific price range.

We quantify the consumption patterns during this period of time. Using a dynamic differences-in-differences model, we find that electricity consumption declined in a sustained manner for several months relative to both the areas with less exposure to foreign electricity markets and to the pre-crisis and pre-subsidy period, indicating a persistent demand response to the high-price environment.

Crucially, while government subsidies provided financial relief, they did not dampen conservation efforts and may have even reinforced them by signaling the severity of the crisis. These findings offer insights for policymakers designing support schemes during energy crises, highlighting the complex interplay between market prices, government interventions, and short-term consumer responses. The analysis of consumers responses in an environment where both the subsidy and a fixed-price regime co-exist is left for future research.

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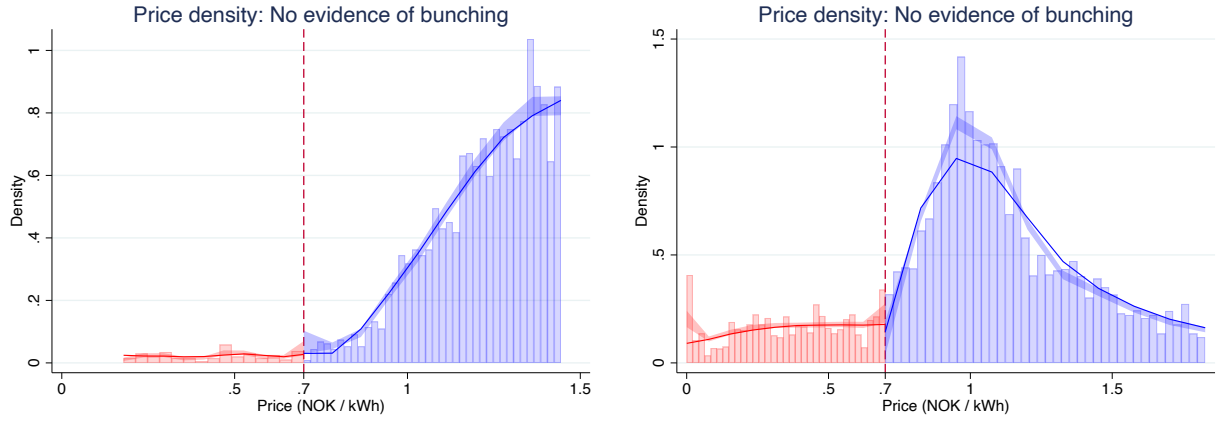
Singhal, P., Kaestner, K., Schwarz, A., and Pahle, M. (2026). How do households react to a real cost shock? Evidence from the energy price crisis in Germany. *Energy Economics*, page 109142.

A Further Details of the RKD Model

A.1 Smoothness of density of the running variable

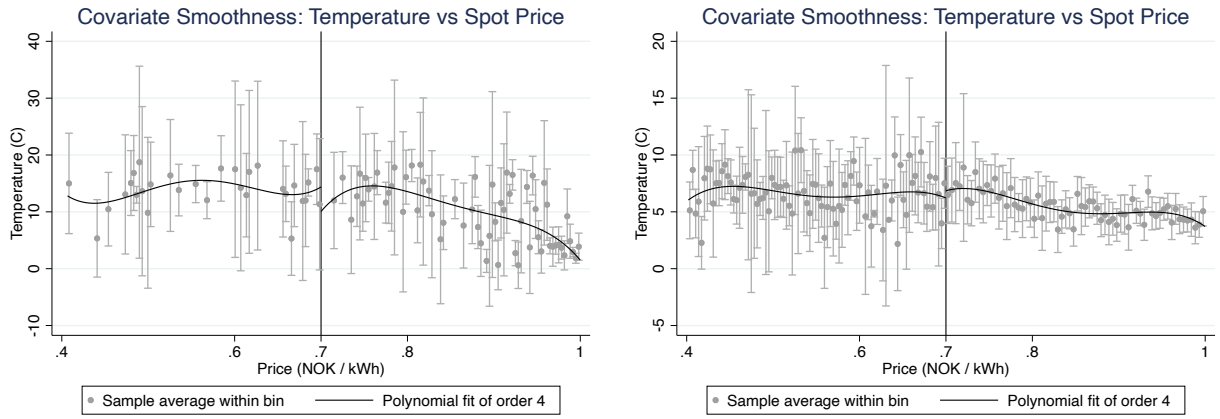
To test the assumption of no precise sorting around the threshold, we implement the robust density test proposed by Cattaneo et al. (2020). We fail to reject the null hypothesis of continuity in the density of the running variable at the 70 øre cutoff (p-value = 0.3764 in Phase 2), suggesting no evidence of threshold manipulation by market participants. In Phase 3, the test gives a lower p-value of 0.0133, but the histogram itself does not show any evidence of bunching.

Figure A.1: Price density. Subsidy Phases: 2 and 3



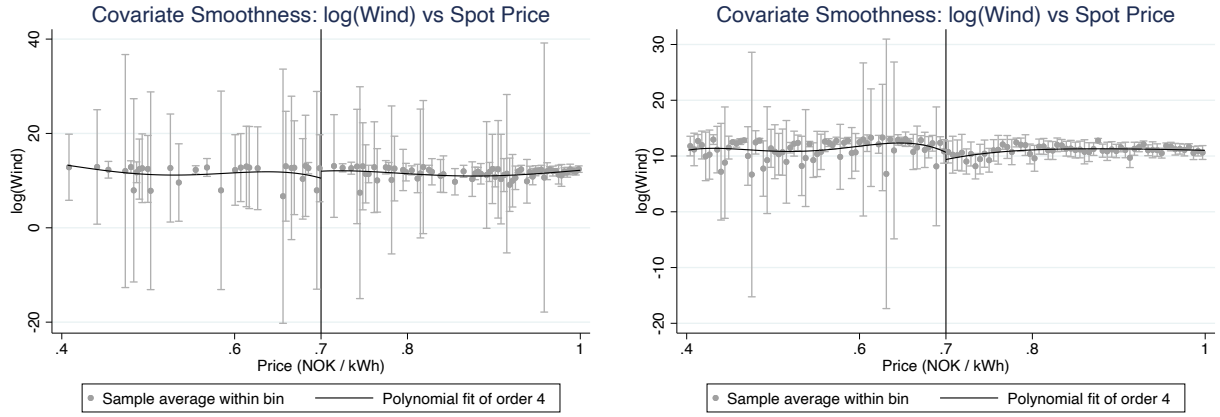
A.2 Continuity of covariates around price threshold

Figure A.2: Temperature and prices. Subsidy Phases 2 and 3



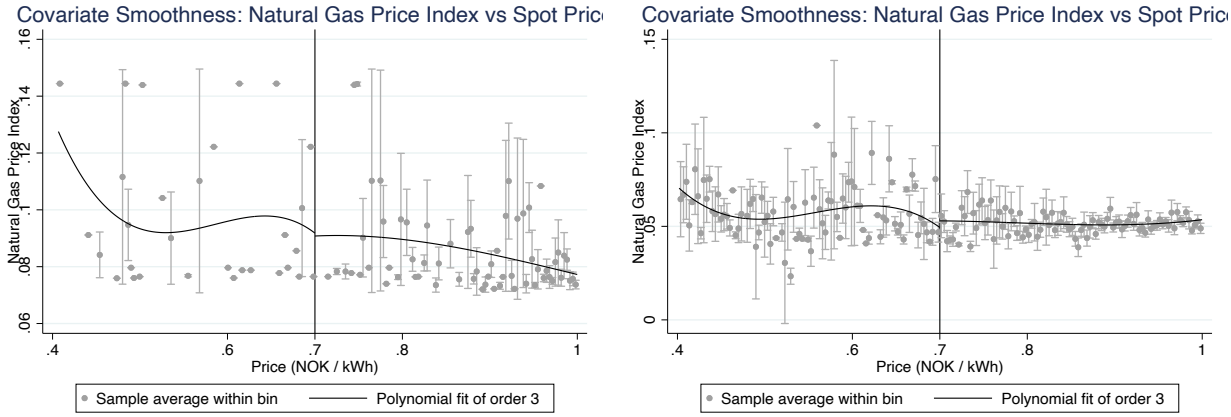
Notes: Binned sample means of temperature and their polynomial fit on each side of the electricity price threshold. Left panel: Phase 2, right panel: Phase 3.

Figure A.3: Wind and prices. Subsidy Phases 2 and 3



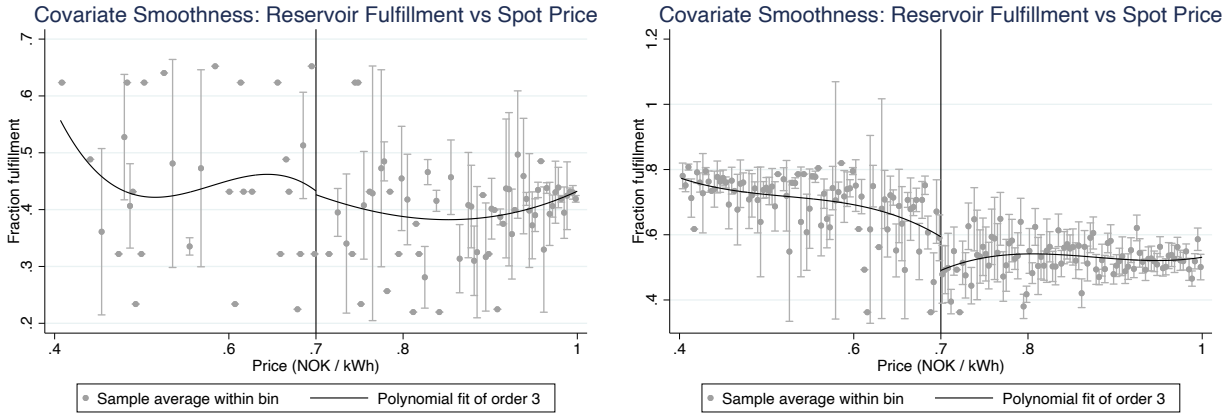
Notes: Binned sample means of log(wind) and their polynomial fit on each side of the electricity price threshold. Left panel: Phase 2, right panel: Phase 3.

Figure A.4: Natural gas and prices. Subsidy Phases 2 and 3



Notes: Binned sample means of the natural gas price index and their polynomial fit on each side of the electricity price threshold. Left panel: Phase 2, right panel: Phase 3.

Figure A.5: Reservoir fulfillment and prices. Subsidy Phases 2 and 3



Notes: Binned sample means of the reservoir fulfillment levels and their polynomial fit on each side of the electricity price threshold. Left panel: Phase 2, right panel: Phase 3.

A.3 Bandwidth

Bandwidth selection in an RKD inherently involves navigating the bias-variance tradeoff. A wider bandwidth increases statistical efficiency by maximizing the number of observations, while a narrower bandwidth reduces bias by restricting the sample to a tighter local neighborhood around the policy threshold.

As a baseline to guide this selection, we estimate the mean squared error (MSE)-optimal bandwidth (Calonico et al. (2014), "CCT" henceforth). To avoid overfitting the local polynomial, this data-driven procedure is implemented without covariates, yielding an optimal bandwidth of 20 øre for Phase 2. In the subsequent estimation step, we incorporate a rich set of fixed effects and non-linear weather controls to increase precision and account for underlying demand shocks. With these controls included, visual inspection of the residualized binscatter reveals a distinct structural break in the slope when employing a slightly wider bandwidth of 30 øre, see Figure A.6.

To rigorously evaluate the stability of our estimates to the choice of bandwidth, we estimate the model using several different bandwidths from 20 øre to 45 øre. At extremely narrow bandwidths, the restricted sample size diminishes statistical power, yielding wide confidence intervals that cover zero. At our preferred bandwidth of 30 øre, however, the 95% confidence interval strictly excludes zero. The point estimates remain statistically significant and relatively stable for bandwidths up to 0.45, though the magnitude attenuates gradually as the window expands. It is worth noting that a bandwidth of 0.45 represents a $\pm 64\%$ deviation relative to the threshold price, whereas a bandwidth of 0.20 constitutes a more localized $\pm 29\%$ deviation. Using our preferred specification with the full set of controls and fixed effects (Column 3 of Table 4), the estimated changes in slope (τ_2) across this grid correspond to implied demand elasticities ranging from -0.707 to -0.227, with an estimated elasticity of -0.634 at the chosen 30 øre bandwidth.

We replicate this procedure for Phase 3. The CCT algorithm suggests an optimal bandwidth of 0.27 øre. However, the estimated changes in slope lose statistical significance at wider bandwidths; consequently, the implied elasticities (evaluated via the Delta method) are also statistically insignificant in that upper range. When controlling for covariates and fixed effects, a slightly narrower bandwidth of 0.20 øre produces a clear structural break in the residualized binscatter (Figure A.6). Relying on this 0.20 øre bandwidth, the implied elasticity is -0.47, with the broader sensitivity analysis bounding the elasticity estimates between -0.87 and -0.09.

Figure A.6: Binscatter of residuals. Subsidy Phase: 2 and 3



Notes: Binscatter of residuals of the RKD regression as a function of the centered spot price for the main specification. The vertical axis is the unexplained variation in consumption after removing out the effects of the covariates and all the fixed effects. The change in slope on each side of the normalized price threshold confirms the validity of the design. Left panel: Phase 2, right panel: Phase 3.

B How Price-Sensitive Are Consumers in Norway?

Electricity plays a uniquely important role in Norwegian households' energy use. According to Statistics Norway, a typical household consumes about 14,900 kWh of electricity annually, equivalent to roughly 1,240 kWh per month, and electricity accounts for nearly 80–90% of total household energy consumption.[28] This share is exceptionally high compared with most European countries, reflecting Norway's extensive electrification of heating and transportation and its reliance on hydropower. Because most homes use electric heating, consumption peaks sharply during the winter months.

A typical electricity bill consists of three roughly equal components: (1) grid rent (*nettleie*), which covers the cost of transporting electricity and maintaining the grid; (2) electricity costs, paid to suppliers based on the spot price and supplier margin; and (3) taxes and fees, including value-added tax (VAT), electricity taxes, and environmental charges. For illustration, a household using 1,300 kWh in a month with an average spot price of 1.20 NOK/kWh would pay around 1,560 NOK for electricity. Adding grid rent (approximately 520 NOK per month) and taxes (roughly 25% VAT plus excise charges) results in a total monthly bill of about 2,600 NOK.[29] Although this varies across bidding zones and seasons, it highlights

[28] SSB, 2024
[29] NVE.no

that electricity spending represents a meaningful share of household budgets—around 5% of disposable income, given a median after-tax income of about 635,000 NOK in 2023.[30]

Because electricity dominates total energy use, fluctuations in electricity prices directly affect households' overall energy expenditures. This makes Norway an especially relevant setting for examining price sensitivity and understanding how consumers adjust their demand in response to changing electricity prices.

However, estimating this sensitivity is empirically challenging because electricity prices are *endogenous* and unobserved regional shocks, such as sudden temperature drops or industrial demand shifts, can affect both consumption and prices, introducing further endogeneity.

To overcome these challenges, we employ a two-stage least squares (2SLS) approach that isolates exogenous price variation arising from the supply side. Specifically, we use the natural gas price index and wind energy generation as instruments for electricity prices. These variables capture cost-side shocks, fluctuations in generation costs and renewable output that influence wholesale electricity prices but are plausibly unrelated to local demand conditions. This instrumental variable strategy allows us to estimate short-run demand elasticities at a high (hourly) frequency while mitigating simultaneity bias. Specifically, we employ the following two-stage least squares (2SLS) approach.

First stage.

$$\log(\text{Price}_{hz}) = \pi_0 + \pi_1 \log(\text{NGas}_{hz}) + \pi_2 \log(\text{WGen}_{hz}) + \mathbf{X}_{hz}\boldsymbol{\pi} + \sigma_{\text{hod}} + \gamma_d + \lambda_m + \mu_y + \nu_{hz}. \quad (3)$$

Second stage.

$$\log(\text{Consumption}_{hz}) = \alpha + \beta \log(\widehat{\text{Price}}_{hz}) + \mathbf{X}_{hz}\boldsymbol{\beta} + \sigma_{\text{hod}} + \gamma_d + \lambda_m + \mu_y + \varepsilon_{hz}. \quad (4)$$

where h indexes *hours* and z indexes zones. $\mathbf{X}_{hz}$ includes temperature^2, heating-degree, and cooling-degree measures at the hourly frequency. We include fixed effects for hour-of-day (σ_{hod}), day-of-week (γ_d), month (λ_m), and year (μ_y). The endogenous variable $\log(\text{Price}_{hz})$ is instrumented with the natural gas price index and wind energy generation. All 2SLS regressions are estimated separately by zone.

The corresponding first-stage results reported in Table B.1 confirm the strong relevance of the instrument, with substantial explanatory power for variation in electricity prices. The Bloomberg EGSI day-ahead natural gas index (EGSIZPAD)—a transparent benchmark built from executed spot trades across European gas hubs and used for EEX settlements—exhibits strong relevance, loading positively on electricity prices in Zones 1, 2, and 5, while small

[30] SSB, 2024

Table B.1: First stage (OLS): Instrument relevance by zone and policy period

	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Panel A: Pre-subsidy period (≤ Nov 2021), raw prices					
log(EGSIZPAD)	0.646^{***}	0.647^{***}	-0.269^{***}	0.016	0.640^{***}
	(0.009)	(0.009)	(0.013)	(0.011)	(0.007)
log(Wind Gen.)	-0.016^{***}	-0.043^{***}	-0.137^{***}	-0.088^{***}	—
	(0.003)	(0.003)	(0.004)	(0.006)	—
F excl. instr.	2,480.28	2,724.75	957.57	92.93	9,605.20
p-value	0.000	0.000	0.000	0.000	0.000
N	7,722	7,989	7,987	7,986	7,991
Panel B: Subsidy period (≥ Dec 2021), raw prices					
log(EGSIZPAD)	1.416^{***}	1.317^{***}	-0.100^{***}	-0.388^{***}	0.970^{***}
	(0.013)	(0.014)	(0.013)	(0.012)	(0.065)
log(Wind Gen.)	-0.020^{***}	-0.093^{***}	-0.313^{***}	-0.351^{***}	0.035^{***}
	(0.005)	(0.005)	(0.007)	(0.008)	(0.010)
F excl. instr.	5,706.62	4,790.87	1,121.82	1,459.35	114.79
p-value	0.000	0.000	0.000	0.000	0.000
N	22,314	22,488	22,482	22,483	4,394
Panel C: Subsidy period (≥ Dec 2021), net prices					
log(EGSIZPAD)	0.928^{***}	0.848^{***}	-0.155^{***}	-0.400^{***}	0.682^{***}
	(0.012)	(0.013)	(0.013)	(0.012)	(0.064)
log(Wind Gen.)	-0.012^{**}	-0.068^{***}	-0.292^{***}	-0.341^{***}	0.034^{***}
	(0.005)	(0.005)	(0.007)	(0.008)	(0.010)
F excl. instr.	2,872.39	2,330.92	1,094.99	1,480.50	60.93
p-value	0.000	0.000	0.000	0.000	0.000
N	22,314	22,488	22,482	22,483	4,394

Notes: The table reports first-stage OLS regressions corresponding to the IV specifications for hourly electricity demand by zone and policy period. The dependent variable in Panels A and B is the log of the raw electricity price; Panel C uses the log of the net-of-subsidy price. The instruments are log natural gas price (EGSIZPAD) and log wind generation. In Panel A, Zone 5, wind generation is excluded because no wind generation observations are available during the pre-subsidy period; natural gas prices therefore serve as the sole instrument in that specification. Dashes indicate unavailable estimates. All regressions control for temperature squared, heating and cooling degree days, and a rich set of time fixed effects. Standard errors in parentheses. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

negative loadings in Zones 3–4 are consistent with their limited exposure to continental gas-driven price formation due to weak interconnection with European markets. Wind generation is price-depressing in Zones 1–4 but slightly price-increasing in Zone 5. Instrument strength is uniformly high as measured by the Kleibergen–Paap F-statistics, mitigating weak-instrument concerns.

Table B.2: Price elasticities of electricity demand by zone and policy period

	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Panel A: Pre-subsidy period (≤ Nov 2021), raw prices					
log(price)	-0.123^{***}	-0.204^{***}	-0.054^{***}	-0.469^{***}	-0.101^{***}
	(0.004)	(0.004)	(0.016)	(0.037)	(0.004)
N	7,722	7,989	7,987	7,986	7,991
Panel B: Subsidy period (≥ Dec 2021), raw prices					
log(price)	-0.026^{***}	-0.025^{***}	-0.063^{***}	-0.020^{***}	-0.022^{***}
	(0.001)	(0.001)	(0.002)	(0.001)	(0.006)
N	22,314	22,488	22,482	22,483	4,394
Panel C: Subsidy period (≥ Dec 2021), net prices					
log(net price)	-0.040^{***}	-0.039^{***}	-0.062^{***}	-0.019^{***}	-0.044^{***}
	(0.001)	(0.002)	(0.002)	(0.001)	(0.009)
N	22,314	22,488	22,482	22,483	4,394

Notes: The table reports IV estimates of the price elasticity of hourly electricity consumption by zone. The dependent variable is the logarithm of hourly electricity consumption. In Panel A, the endogenous regressor is the logarithm of electricity prices during the pre-subsidy period. In Panel B, the endogenous regressor is the logarithm of electricity prices during the subsidy period. In Panel C, electricity prices are replaced by the net price that accounts for the subsidy scheme. Electricity prices are instrumented using gas prices and wind generation. All specifications include temperature controls (HDD, CDD, and squared temperature), as well as hour-of-day and day-of-week fixed effects. Month and year fixed effects are included in Panels B and C. Standard errors are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Next, Table B.2 reports hourly price elasticity estimates by zone across three specifications: the pre-subsidy period using raw prices (Panel A), the subsidy period using raw prices (Panel B), and the subsidy period using net consumer prices that account for the subsidy (Panel C). All elasticities are estimated via two-stage least squares regressions with two instrumental variables, as described above.

Before the introduction of the subsidy, estimated price elasticities ranged from -0.05 to -0.47. Following the introduction of the subsidy, electricity consumption declined as shown in Figure 7. This suggests that consumers responded to high electricity prices, even though

the subsidy covered 80–90% of the price exceeding the specified threshold. One possible explanation is that the threshold remained relatively high, meaning that the effective price faced by consumers was still substantially above pre-crisis levels.

C Differences-in-Differences Analysis

We estimate a differences-in-differences (DiD) specification that exploits regional variation in exposure to the European Energy Crisis. Zones that are largely independent of electricity flows with the rest of Europe serve as the control group, while southern zones that are physically interconnected with other European electricity markets constitute the treated group. This framework estimates the treatment effect of the net price increase on electricity consumption, where the net price encompasses both the Energy Crisis shock and the subsidy. The parallel trends assumption is discussed in subsection 5.1 and in addition we also conduct a formal test of the parallel-trends assumption using only the clean pre-shock period, January–August 2021 inspired by the work in Brolinson et al. (2026) that explain below. We emphasize that this analysis does not capture the causal effect of the subsidy alone on consumption.

The empirical model is given by:

$$\log Y_{izt} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Treated}_z + \beta_3 (\text{Post}_t \times \text{Treated}_z) + \mathbf{X}_{zt}\boldsymbol{\beta} + \delta_i + \alpha_z + \gamma_m + \eta_y + \varepsilon_{izt}, \quad (5)$$

where Y_{izt} denotes average electricity consumption of household i in zone z and month t. Post_t is an indicator equal to one for periods after December 2021, marking the introduction of the electricity subsidy scheme. Treated_z is an indicator equal to one for zones exposed to the electricity price increase (Zones NO1, NO2, and NO5). The interaction term ($\text{Post}_t \times \text{Treated}_z$) captures the differences-in-differences estimate of the treatment effect, and β_3 is therefore the main coefficient of interest. $\mathbf{X}_{zt}$ is a vector of weather-related control variables: temperature, temperature squared, HDD, and CDD. Household fixed effects are denoted by δ_i, zone fixed effects by α_z, month fixed effects by γ_m, and year fixed effects by η_y.[31] The error term ε_{izt} captures idiosyncratic shocks at the household–zone–time level.

The validity of the differences-in-differences framework relies on the assumption that treated and control zones would have followed parallel trends in the absence of treatment. To

[31] We include separate month and year fixed effects (i.month and i.year), rather than a full set of month–year fixed effects in order to capture seasonal patterns. Since the sample spans January 2021 to December 2023, the post-policy indicator varies within both month and year dimensions and is therefore not absorbed by these fixed effects. However, see robustness check on this structure of time fixed effects in Table C.5.

assess this assumption, we provide three arguments. First, Figure 2 plots mean log electricity consumption by zone. The figure shows broadly similar consumption patterns prior to the onset of the subsidy. In particular, electricity consumption declines from January 2021 to January 2022 in bidding zones NO1, NO2, and NO5, while consumption in NO3 and NO4 does not exhibit a comparable reduction over the same period.

Second, the dynamic effects model of subsection 5.1 shows that the two groups do not exhibit material differences in consumption in the pre-period. The discussion there expands on this argument.

Third, following Brolinson et al. (2026), we interact the high-price-zone indicator with a linear monthly time trend. The estimated coefficient is -0.01 (s.e. $= 0.019$; $p = 0.52$) and is statistically indistinguishable from zero. We therefore find no evidence that electricity consumption in the high- and low-price zones followed different trends prior to the electricity-price shock.

C.1 Baseline differences-in-differences results

This subsection presents the baseline differences-in-differences estimates from Equation 5, where December 2021 is defined as the start of the treatment period. Table C.1 reports the results using data from January 2021 through December 2023. Across all specifications, the coefficient on the interaction term ($Treated \times Post$) is negative and statistically significant, indicating a reduction in electricity consumption in treated zones relative to control zones. The estimated effect is approximately 8 percent in most specifications, with the magnitude remaining stable after the inclusion of fixed effects and controls.[32] This result is consistent with evidence that household electricity demand may respond to salient policy interventions and public signals, in addition to standard marginal price incentives (Reiss and White, 2008; Jacobsen et al., 2012).[33]

The results also account for weather conditions. In columns (4)–(5), the inclusion of weather-related control variables shows that the treatment effect remains robust to the inclusion of these controls, with only a modest reduction in magnitude (from –0.086 to approximately –0.07).

[32] Using the coefficients from Table C.1 the exact percentage change is between $100 \times (\exp(-0.07) - 1) = -6.76\%$ and $100 \times (\exp(-0.087) - 1) = -8.33\%$.

[33] Our findings are also related to evidence found using quasi-experimental data that consumers place substantial weight on persistent electricity price components when forming expectations and making long-term energy decisions, such as heating technology investments (Sahari (2019)).

Table C.1: Differences-in-differences estimates

	(1)	(2)	(3)	(4)	(5)
Treated × Post	-0.087***	-0.087***	-0.087***	-0.070***	-0.068***
	(0.012)	(0.012)	(0.012)	(0.013)	(0.013)
Post	0.0003	-0.086***	0.0002	-0.102***	-0.104***
	(0.002)	(0.012)	(0.002)	(0.024)	(0.024)
Treated	-0.090	-0.090			
	(0.074)	(0.074)			
Temperature				-0.023**	-0.023**
				(0.012)	(0.012)
Temperature^2				0.0006***	0.0006***
				(0.0002)	(0.0002)
Heating Degree Days				-0.007	-0.007
				(0.015)	(0.015)
Cooling Degree Days				-0.027	-0.027
				(0.025)	(0.025)
Intercept	7.204***	7.744***	7.136***	7.780***	7.790***
	(0.072)	(0.081)	(0.008)	(0.254)	(0.254)
Zone FE	No	No	Yes	Yes	Yes
Month & Year FE	No	Yes	No	Yes	Yes
Household FE	No	No	No	No	Yes
DV mean	7.107	7.107	7.107	7.107	7.107
R^2	0.005	0.602	0.005	0.606	0.606
N	52,783,891	52,783,891	52,783,891	52,783,891	52,783,891

Notes: The dependent variable is the logarithm of monthly average electricity consumption per meter (household). The sample covers the period from January 2021 to December 2023. Zones 1, 2, and 5 are treated, while Zones 3 and 4 serve as control zones. Columns (1)–(3) present baseline difference-in-differences specifications with alternative combinations of zone and time fixed effects. Column (4) augments the specification with weather controls (temperature, temperature squared, heating degree days, and cooling degree days). Column (5) further includes household fixed effects, absorbing time-invariant heterogeneity at the household level. Standard errors clustered at the electricity-zone level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The post-treatment indicator (*Post*) is small and statistically insignificant in specifications without time fixed effects, indicating no uniform shift in consumption across zones after the treatment date. Once month and year fixed effects are included, however, the coefficient on *Post* becomes negative and statistically significant, reflecting the absorption of common aggregate time shocks that would otherwise be attributed to the post period. Similarly, the treated-zone indicator (*Treated*) is negative and statistically significant in specifications without zone fixed effects, indicating lower baseline consumption levels in treated zones relative to control zones. Once zone fixed effects are included, this indicator is absorbed and

omitted from the regression, as it is time-invariant at the zone level.

Our findings confirm that the price differential stemming from the difference in the degree of interconnection for the two groups with the rest of Europe has quantifiable and economically important effects on electricity consumption.

C.2 Robustness checks: alternative control groups and time spans

Alternative control groups. We test whether the results are sensitive to the choice of the control group by re-estimating the differences-in-differences model using each control zone separately rather than pooling Zones 3 and 4.

Table C.2 reports the results when Zone 3 alone is used as the control group, while Table C.3 shows the corresponding results using Zone 4 only. In both cases, the dependent variable is the log of monthly electricity consumption per household. The treatment effect remains negative and statistically significant, confirming that the observed demand reduction in treated zones is not driven by the composition of the control group. The magnitude of the effect is somewhat larger when Zone 4 (the northern zone with higher electricity consumption and colder climate) is used as the control, while Zone 3 yields smaller estimates. This outcome is not surprising, as demonstrated in Figure 2, Zone 4 experienced little or no price increase during the critical period of 2022–2023, whereas prices rose sharply in the treated zones, making Zone 4 a more comparable benchmark for identifying demand responses. It is not possible to compute standard errors clustered at the electricity zone level since the control group contains only one zone in each of these regressions.

Keeping data up to August 2022: Before the subsidy coverage changed from 80 to 90%. Table C.4 reports results similar to those in Table C.1 but restricting the sample up to August 2022, which is the last month before the subsidy coverage changed from 80 to 90%. This addresses the potential concern that consumers' behavior changed once they were offered a slightly more generous subsidy. Using this restricted sample the coefficients vary between -0.081 to -0.097 across the different specifications and remain statistically significant.

Control group choice and time span. To assess simultaneously the influence of the choice of the control zone and the change in the amount of the subsidy, we re-estimate the model using data up to August 2022 and using as control group only zone 3 or zone 4,

Table C.2: DiD estimates of monthly household electricity consumption
(Zone 3 as control)

	(1)	(2)	(3)	(4)	(5)
Treated × Post	-0.089***	-0.089***	-0.089***	-0.060***	-0.058***
	(0.0006)	(0.0005)	(0.0006)	(0.0005)	(0.0003)
Post	0.002***	-0.090***	0.002***	-0.116***	-0.117***
	(0.0005)	(0.0008)	(0.0005)	(0.0008)	(0.0004)
Treated	-0.012***	-0.012***			
	(0.0005)	(0.0004)			
Temperature				-0.027***	-0.027***
				(0.0003)	(0.0001)
Temperature^2				0.0008***	0.0008***
				(4×10^{-6})	(2×10^{-6})
Heating Degree Days				-0.009***	-0.009***
				(0.0003)	(0.0001)
Cooling Degree Days				-0.033***	-0.033***
				(0.0004)	(0.0002)
Intercept	7.127***	7.678***	7.136***	7.815***	7.807***
	(0.0004)	(0.0005)	(0.0003)	(0.005)	(0.002)
Zone FE	No	No	Yes	Yes	Yes
Month & Year FE	No	Yes	No	Yes	Yes
Household FE	No	No	No	No	Yes
DV mean	7.078	7.078	7.078	7.078	7.078
Adj.R^2	0.005	0.603	0.005	0.610	0.607
N	46,680,509	46,680,509	46,680,509	46,680,509	46,680,509

Notes: The dependent variable is the logarithm of monthly average electricity consumption per meter (household). The sample covers the period from January 2021 to December 2023. Zones 1, 2, and 5 are treated, while Zone 3 serves as control zone. Columns (1)–(3) report baseline difference-in-differences specifications with alternative combinations of zone and time fixed effects. Column (4) augments the specification with weather controls, including temperature, temperature squared, heating degree days, and cooling degree days. Column (5) further includes household fixed effects, absorbing time-invariant heterogeneity at the household level. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.3: DiD estimates of monthly household electricity consumption
(Zone 4 as control)

	(1)	(2)	(3)	(4)	(5)
Treated × Post	-0.083***	-0.083***	-0.083***	-0.084***	-0.081***
	(0.0007)	(0.0006)	(0.0007)	(0.0006)	(0.0003)
Post	-0.003***	-0.083***	-0.003***	-0.097***	-0.10***
	(0.0006)	(0.0009)	(0.0006)	(0.0009)	(0.0004)
Treated	-0.203***	-0.203***			
	(0.0006)	(0.0005)			
Temperature				-0.013***	-0.013***
				(0.0003)	(0.0002)
Temperature^2				0.0007***	0.0007***
				(4×10^{-6})	(2×10^{-6})
Heating Degree Days				0.005***	0.005***
				(0.0003)	(0.0002)
Cooling Degree Days				-0.036***	-0.036***
				(0.0004)	(0.0001)
Intercept	7.318***	7.872***	7.136***	7.563**	7.573**
	(0.0005)	(0.0005)	(0.0002)	(0.006)	(0.003)
Zone FE	No	No	Yes	Yes	Yes
Month & Year FE	No	Yes	No	Yes	Yes
Household FE	No	No	No	No	Yes
DV mean	7.10	7.101	7.101	7.10	7.101
Adj.R^2	0.005	0.611	0.005	0.620	0.620
N	43,879,308	43,879,308	43,879,308	43,879,308	43,879,308

Notes: The dependent variable is the logarithm of monthly average electricity consumption per household. The sample covers the period from January 2021 to December 2023. Treatment starts in December 2021. Zones 1, 2, and 5 are treated, while Zone 4 serves as the control group. Columns (1)–(3) report baseline difference-in-differences specifications with alternative combinations of zone and time fixed effects. Column (4) augments the specification with weather controls, including temperature, temperature squared, heating degree days, and cooling degree days. Column (5) further includes household fixed effects, absorbing time-invariant heterogeneity at the household level. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

respectively. In both cases, the coefficient remains stable and statistically significant at the level of the previous paragraph where we used as control group zones 3 and 4 together. These results are available upon request.

Alternative fixed-effects structure. Table C.5 shows the results from the main differences-in-differences specification using month-year fixed effects instead of month and year fixed-effects. Only columns (2), (4), and (5) are affected by this change since the other columns do not include time fixed-effects. We present them nonetheless for completeness. The coefficient of the interaction in those columns changed from -0.087, -0.070, and -0.068 to -0.087, -0.079, and -0.077, respectively. Therefore, our results are robust to this change in the time fixed effects structure.

Table C.4: DiD estimates of monthly household electricity consumption (sample up to August 2022)

	(1)	(2)	(3)	(4)	(5)
Treated × Post	-0.097***	-0.097***	-0.097***	-0.082***	-0.081***
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Post	-0.006**		-0.006**		-0.092***
	(0.003)		(0.003)		(0.004)
Treated	-0.090	-0.090			
	(0.074)	(0.074)			
Temperature				-0.023*	-0.023*
				(0.014)	(0.014)
Temperature^2				0.0004	0.0004
				(0.0004)	(0.0004)
Heating Degree Days				-0.011	-0.010
				(0.020)	(0.020)
Cooling Degree Days				-0.025	-0.025
				(0.035)	(0.035)
Intercept	7.204***	7.786***	7.128***	7.860***	7.874***
	(0.073)	(0.082)	(0.003)	(0.348)	(0.347)
Zone FE	No	No	Yes	Yes	Yes
Month & Year FE	No	Yes	No	Yes	Yes
Household FE	No	No	No	No	Yes
DV mean	7.088	7.088	7.088	7.088	7.088
Adj.R^2	0.007	0.660	0.008	0.662	0.662
N	29,473,953	29,473,953	29,473,953	29,473,953	29,473,953

Notes: The dependent variable is the logarithm of monthly average electricity consumption per household. The sample covers the period from January 2021 through August 2022. Treatment starts in December 2021. Zones 1, 2, and 5 are treated, while Zones 3 and 4 serve as control zones. Columns (1)–(3) report baseline difference-in-differences specifications with alternative combinations of zone and time fixed effects. Column (4) augments the specification with weather controls, including temperature, temperature squared, heating degree days, and cooling degree days. Column (5) further includes household fixed effects, absorbing time-invariant heterogeneity at the household level. Standard errors clustered at the electricity-zone level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.3 Heterogeneous effects

Table C.6 reports the full set of results for the households characteristics and electricity consumption. The coefficients on Treated × Post × Group j, $j = 2, \ldots, 5$, are the heterogeneity effects reported in Table 5.

Table C.5: Differences-in-differences estimates of household electricity consumption with month-year FEs

	(1)	(2)	(3)	(4)	(5)
Treated × Post	-0.087***	-0.087***	-0.087***	-0.079***	-0.077***
	(0.0005)	(0.0004)	(0.0005)	(0.0004)	(0.0002)
Post	0.0003		0.0002		
	(0.0004)		(0.0004)		
Treated	-0.090***	-0.090***			
	(0.0003)	(0.0004)			
Temperature				0.007***	0.007***
				(0.0004)	(0.0002)
Temperature^2				0.00003***	0.00004***
				(4×10^{-6})	(2×10^{-6})
Heating Degree Days				0.012***	0.012***
				(0.0004)	(0.0002)
Cooling Degree Days				-0.023***	-0.024***
				(0.0004)	(0.0002)
Intercept	7.204***	7.844***	7.136***	7.566***	7.572***
	(0.0003)	(0.0006)	(0.0002)	(0.0062)	(0.0029)
Zone FE	No	No	Yes	Yes	Yes
Month-Year FE	No	Yes	No	Yes	Yes
Household FE	No	No	No	No	Yes
DV mean	7.107	7.107	7.107	7.107	7.107
Adj. R^2	0.005	0.615	0.005	0.617	0.617
N	52,783,891	52,783,891	52,783,891	52,783,891	52,783,891

Notes: The dependent variable is the logarithm of monthly average electricity consumption per meter (household). The sample covers the period from January 2021 to December 2023. Zones 1, 2, and 5 are treated, while Zones 3 and 4 serve as control zones. Columns (1) and (3) report the baseline difference-in-differences specifications. Columns (2), (4), and (5) use month-year fixed effects, allowing common time shocks to vary flexibly across each month-year combination. Columns (4) and (5) additionally include weather controls (temperature, temperature squared, heating degree days, and cooling degree days). Column (5) further includes household fixed effects, absorbing time-invariant heterogeneity at the household level. Standard errors are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C.6: Full regression results: heterogeneous treatment effects

	Dependent variable: Log electricity consumption				
	Income (1)	Usable floor area (2)	Building vintage (3)	Number of rooms (4)	Household size (5)
Baseline treatment effects					
Post	−0.111***	−0.111***	−0.112***	−0.111***	−0.101***
	(0.024)	(0.026)	(0.023)	(0.024)	(0.024)
Treated × Post	−0.055***	−0.054***	−0.071***	−0.053***	−0.058***
	(0.012)	(0.014)	(0.013)	(0.011)	(0.012)
Post × household-characteristic group					
Post × Group 2	0.0053***	0.0012***	0.0088***	0.0881***	−0.006***
	(0.0006)	(0.008)	(0.0012)	(0.0019)	(0.0004)
Post × Group 3	0.0086***	0.005***	0.0088***	0.0085***	−0.0068***
	(0.0015)	(0.006)	(0.0001)	(0.0011)	(0.0008)
Post × Group 4	0.015***	0.011***	0.022***	0.0047***	−0.0025**
	(0.001)	(0.005)	(0.001)	(0.0023)	(0.0011)
Post × Group 5	0.011***	0.009***	0.037***	0.0067***	0.0013
	(0.002)	(0.007)	(0.005)	(0.0047)	(0.0018)
Heterogeneity estimates					
Treated × Post × Group 2	−0.011***	−0.0017	−0.0003	−0.0138***	−0.0119***
	(0.0016)	(0.0084)	(0.0013)	(0.0021)	(0.0033)
Treated × Post × Group 3	−0.017***	−0.0104	0.0053***	−0.0162***	−0.0153***
	(0.0024)	(0.0071)	(0.001)	(0.0042)	(0.0021)
Treated × Post × Group 4	−0.0204***	−0.0208***	0.0201***	−0.0195***	−0.0148***
	(0.0031)	(0.0082)	(0.0017)	(0.0083)	(0.0035)
Treated × Post × Group 5	−0.017***	−0.0249**	0.0275***	−0.0265***	−0.0163***
	(0.0056)	(0.0118)	(0.0063)	(0.0068)	(0.0035)
Weather controls					
Temperature	−0.0231**	−0.0231**	−0.0231**	−0.0228**	−0.0231**
	(0.0118)	(0.0118)	(0.0188)	(0.0118)	(0.0118)
Temperature^2	0.0006***	0.0006***	0.0006***	0.0006***	0.0006***
	(0.0002)	(0.0003)	(0.0003)	(0.0003)	(0.0003)
Heating Degree Days	−0.0075	−0.0075	−0.0074	−0.0071	−0.0075
	(0.0153)	(0.0153)	(0.0153)	(0.0154)	(0.0154)
Cooling Degree Days	−0.0273	−0.0275	−0.0276	−0.0282	−0.0275
	(0.0251)	(0.0251)	(0.0251)	(0.0251)	(0.0251)
Intercept	7.793***	7.792***	7.791***	7.789***	7.791***
	(0.253)	(0.252)	(0.253)	(0.254)	(0.254)
Zone FE	Yes	Yes	Yes	Yes	Yes
Month & Year FE	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes
Observations	52,244,018	52,097,156	51,642,742	48,830,338	52,783,891

Notes: The dependent variable is the logarithm of monthly household electricity consumption. Each column reports a separate household fixed-effects regression. Column (1) estimates heterogeneity by household income, Column (2) by usable floor area, Column (3) by building vintage, Column (4) by number of rooms, and Column (5) by household size. In each specification, Group 1 of the corresponding household characteristic is the omitted reference category. Post is an indicator for the period after the introduction of the electricity subsidy scheme, while Treated identifies households located in Zones NO1, NO2, and NO5. All specifications include household, zone, month, and year fixed effects and weather controls. The number of observations varies across columns due to differences in data availability for the household characteristic used to construct the heterogeneity groups. See Table C.7 for the construction of household characteristic groups. Standard errors are clustered at the zone level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table C.7: Construction of household characteristic groups and summary statistics

Variable (Norwegian name)	Group	Definition	N
Household income (husholdningsinntekt, annual NOK)			
	1	Lowest income quintile: $<$ 407,635	298,727
	2	407,635 – 609,894	510,336
	3	609,894 – 831,033	719,585
	4	831,033 – 1,096,594	953,284
	5	Highest income quintile: $\geq$ 1,096,594	1,531,810
Dwelling size (bruksareal, m^2)			
	1	Less than 50 m^2	869,001
	2	50–99 m^2	7,887,352
	3	100–149 m^2	13,771,869
	4	150–199 m^2	12,511,329
	5	200 m^2 or more	11,844,832
Construction period (byggeår)			
	1	Built before 1960	12,454,449
	2	Built 1960–1989	20,660,649
	3	Built 1990–2009	8,995,821
	4	Built 2010–2019	4,148,709
	5	Built 2020 or later	269,212
Number of rooms (antall rom)			
	1	1–2 rooms	3,155,721
	2	3–4 rooms	18,702,429
	3	5–6 rooms	16,682,043
	4	7–8 rooms	4,419,969
	5	9 rooms or more	996,123
Household size (antall personer)			
	1	1 person	1,201,641
	2	2 persons	1,521,057
	3	3 persons	877,562
	4	4 persons	1,127,399
	5	5 persons or more	619,757

Notes: Household income is grouped into quintiles based on the sample distribution; reported figures are mean income (NOK) within each quintile. All other variables are grouped using fixed thresholds, and reported figures are numbers of household–month observations. Group 1 is the omitted reference category in the heterogeneous treatment effect regressions.